OSHAcademy™
Occupational Safety & Health Training

Fall Protection in Construction

Elements for 29 CFR 1926.500-503

In the U.S. construction industry, falls are the leading cause of worker fatalities and one of OSHA's most commonly cited violations. On average, between 150 and 200 workers are killed and more than 100,000 are injured as a result of falls at construction sites each year. This course looks at how to identify fall hazards, fall protection systems, the importance of training, inspection and maintenance of equipment, and rescuing at height.

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OSHAcademy Course 805 Study Guide

Fall Protection in Construction

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Contact OSHAcademy to arrange for use as a training document.

This study guide is designed to be reviewed off-line as a tool for preparation to successfully complete OSHAcademy Course 805.

Read each module, answer the quiz questions, and submit the quiz questions online through the course webpage. You can print the post-quiz response screen which will contain the correct answers to the questions.

The final exam will consist of questions developed from the course content and module quizzes.

We hope you enjoy the course and if you have any questions, feel free to email or call:

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Course Introduction

Go to any construction site and watch those who are working above a lower level. Should they be wearing fall protection? Should they be protected by fall protection systems? Should they be using fall-protection methods? Do they need fall protection training? Are they following fall protection rules? Fall protection is a concept that's hard to describe. Ask 10 people what fall protection means and you're likely to get 10 different answers. Is it possible to make sense of fall protection? We think so.

It's important that you be familiar with OSHA's fall protection standards to help save lives and avoid OSHA citations.

Take a look at OSHA's top 10 most cited violations for Fiscal Year 2020 and you will see that fall protection ranks as the number one most commonly cited violation!

OSHA Standard for Fall Protection

In the construction industry in the U.S., falls are the leading cause of worker fatalities. Each year, on average, between 150 and 200 workers are killed and more than 100,000 are injured as a result of falls at construction sites. OSHA recognizes that accidents involving falls are generally complex events frequently involving a variety of factors. Consequently, the standard for fall protection deals with both the human and equipment related issues in protecting workers from fall hazards.

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OSHA has revised its construction industry safety standards and developed systems and procedures designed to prevent employees from falling off, onto, or through working levels and to protect employees from being struck by falling objects. The performance-oriented requirements make it easier for employers to provide the necessary protection. The rule covers most construction workers except those inspecting, investigating, or assessing construction worksite conditions prior to the actual start of work or after all work has been completed.

The rule identifies areas or activities where fall protection is needed. These include, but are not limited to:

- ramps
- runways
- other walkways
- excavations
- hoist areas
- holes
- formwork and reinforcing steel
- leading edge work
- unprotected sides and edges
- overhand bricklaying and related work
- roofing work
- precast concrete erection
- wall openings

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- residential construction
- other walking/working surfaces

The rule sets a uniform threshold height of 6 feet (1.8 meters), thereby providing consistent protection. This means that construction employers must protect their employees from fall hazards and falling objects whenever an affected employee is 6 feet (1.8 meters) or more above a lower level. Protection also must be provided for construction workers who are exposed to the hazard of falling into dangerous equipment.

Under the standard, employers are able to select fall protection measures compatible with the type of work being performed. Fall protection generally can be provided through the use of guardrail systems, safety net systems, personal fall arrest systems, positioning device systems, and warning line systems, among others.

The OSHA rule clarifies what an employer must do to provide fall protection for employees, such as identifying and evaluating fall hazards and providing specific training. Requirements to provide fall protection for workers on scaffolds and ladders and for workers engaged in steel erection of buildings are covered in other subparts of OSHA regulations.

This course is intended primarily for construction-industry employers, employees, and others who don't have a professional background in fall protection and who want to see the "big picture." Generally, the course covers the safe practices in 29 CFR 1926, the primary fall protection rules for construction-industry employers. To complete the picture, the course also highlights fall protection requirements for work on ladders and scaffolds.

This course is organized so you can read each module in the order presented or move about as you choose.

Credits:

Special thanks to Oregon OSHA Technical Resources for producing the primary source document for this training, Fall Protection for the Construction Industry.

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Module 1: Falling: Truths and Consequences

Introduction

Ever since we began working above ground level, falls have been among the most common causes of serious injuries and deaths. We all need fall protection because even those of us with years of experience working at heights can suddenly lose our balance or grip and fall. We may think that our reflexes are fast enough to protect us, but even if we have great balancing and motor skills, and when we're distracted, we're at risk of falling.

We've been falling since Day One. Until we get better at landing, we'll need protection from falling.

And, as the photo to the right illustrates, some workers just don't understand fall hazards. Consequently, they do things at work and home that defy good sense. The information in this course will help alert and prevent workers from falls.

Quiz Instructions

After each section, there is a quiz question. Make sure to read the material in each section to discover the correct answer to these questions. Circle the correct answer. When you are finished go online to take the final exam. This exam is open book, so you can use this study guide.

1. Why does every employee working at heights need fall protection?

a. To reduce the probability of a fall

b. Even experienced workers can fall

It provides a sense of security

To comply with OSHA requirements

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Falling Causes

Construction is a potentially high hazard industry for those who work in it, with falls at the top of the hazards list. Falls are the most frequent cause of fatalities at construction sites and annually account for one of every three construction-related deaths. Here are some interesting statistics:

- Each year in the U.S. more than 300 construction workers are killed and more than 10,000 are seriously injured by falls from heights.
- According to the Bureau of Labor Statistics (BLS), slips, trips, or falls took the lives of 887 workers in 2017 - the highest number in over 30 years.
- Falls from roofs are the most frequent cause of fatal falls in construction, and 80% of rooftop falls are fatal.

Below is a prioritized list showing the types of falls that cause the most non-fatal injuries. As you can see, most fall injuries are caused by falls from ladders.

1. Falls from ladders
2. Falls to lower level, unspecified
3. Falls from roofs
4. From scaffolds or staging
5. Falls from non-moving vehicles
6. Falls from floors, docks, or ground level
7. Falls down stairs
8. Falls from girders or structural steel
9. Falls from piled or stacked material

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2. Which type of fall results in the most construction industry fatalities?

a. Falls off ladders
b. Falls off roofs
c. Falls from scaffolds
d. Falls downstairs

Fall Prevention and Protection

For many in the construction industry, when speaking of fall-protection, equipment is the first thing that comes to mind: personal fall-arrest systems, safety nets, or guardrails, for example. But fall protection means more than equipment. Fall protection is what you do to eliminate fall hazards, prevent falls, and ensure that workers who may fall aren't injured.

Although fall hazards are common at construction worksites, fall-related injuries and fatalities are preventable. Fall hazards can be addressed in two main ways:

1. **Preventing falls:** Workers prevent falls by eliminating the need to work above ground level and by using passive fall prevention controls such as using guardrails, positioning and restraint systems, and hole covers.

2. **Limiting falls:** Workers limit injury after a fall by using active fall protection such as personal fall arrest systems (PFAS) or safety nets and having an effective rescue plan.

To prevent falls, be sure to do the following:

- make fall protection part of your construction worksite safety and health program;
- identify and evaluate fall hazards before the start of work;
- eliminate fall hazards, if possible;
- train workers to recognize and control fall hazards;
- use appropriate equipment to prevent falls and to protect workers if they do fall;
- inspect and maintain fall-protection equipment before and after using it; and
- become familiar with OSHA and company fall-protection guidelines and mandatory requirements.

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3. Which of the following methods is an example of fall prevention?

a. Safety nets
b. Fall arrest systems
c. Guardrails
d. A rescue plan

Fall Protection Hierarchy

It's important to use a systematic "Hierarchy of Controls" for fall protection. In descending order of preference, the hierarchy of controls for fall protection is as follows:

- Elimination or substitution. Examples: Eliminate a hazard by lowering the work surface to ground level. Substitute a hazard by moving a process, sequence, or procedure to a different location so that workers no longer approach a fall hazard.

- Passive fall protection. Isolate or separate the hazard or work practice from workers using guardrails or covers over exposed floor openings.

- Fall restraint. Secure the worker to an anchor using a lanyard short enough to prevent the worker's center of mass from reaching the fall hazard.

- Fall arrest. This includes systems designed to stop a worker's fall after a fall has begun.

- Administrative controls. These work practices or procedures signal or warn a worker to avoid approaching a fall hazard. For example, a person may be appointed to monitor work around fall hazards.

4. When possible, what is the most effective hazard control during construction?

a. Elimination
b. Passive fall protection
c. Fall restraint
d. Administrative controls

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Fall Protection Rules

Everyone in the construction worksite has a role to play in preventing falls.

- **Employers***:* Identify fall hazards at the site. Eliminate the hazards, prevent falls from occurring, or ensure that if falls occur, employees aren't injured. Make sure that employees follow safe practices, use fall protection equipment properly, and are trained to recognize fall hazards.

- **Employees***:* Follow safe work practices, use equipment properly, and participate in training. Learn to recognize unsafe practices, know the tasks that increase the risk of falling, and understand how to control exposure to fall hazards.

- **Architects and engineers***:* Educate employers about hazards that could expose workers to falls during each phase of the project. When designing buildings and structures, consider fall protection and other safety needs of those who will do the construction work.

- **Building owners and managers***:* Ensure that those who do exterior construction or maintenance work know how to protect themselves from falls, are aware of installed anchorages, and know how to use their fall protection equipment.

- **Equipment manufacturers***:* Ensure that fall protection equipment meets federal OSHA and ANSI safety requirements and protects workers when they use it properly. Warn workers through instruction manuals and on equipment labels about the danger of using equipment improperly.

- **Lawyers***:* Review your client's construction bids to ensure that they comply with OSHA requirements. The documents should clearly state the client's responsibilities for protecting workers from falls and for identifying and controlling hazards that cause falls.

5. Who is responsible for recognizing and eliminating fall hazards, ensuring training is conducted, and preventing injuries?

a. Manufacturers
b. Employees
c. Employers
d. Building owners

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Scenario

On a Friday in June, an estimator arrived at a remodel job to look at a cedar-shake roof and estimate the cost of an addition that a construction crew was building. He spoke to the supervisor at the site and climbed to the roof through an open skylight, using a metal extension ladder.

However, he was unaware that the contractor had used a sheet of thin insulating material to cover three 2-by-6-foot skylight openings in the roof (it had rained the day before). He stepped onto the insulating material, fell through one of the skylights, and landed on his back, 15 feet below.

The supervisor and two subcontractors heard the estimator fall and rushed to the accident. One of the subcontractors used his cell phone to call emergency medical services. EMTs arrived about five minutes later, stabilized the victim and took him to a hospital where he underwent emergency surgery for spinal injuries.

Findings: The employer failed to properly cover the skylight openings on the roof or warn workers about the hazard.

6. Which of the following would fail to provide adequate protection against falls through skylights?

a. Screens
b. Glass/plastic covers
c. Railings or guardrails
d. Fall arrest systems

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Module 2: Preparing to Prevent Falls

Fall Protection Program

A construction fall protection program is what you and your coworkers do to achieve and maintain a safe, healthful construction worksite. There are as many types of safety and health programs as there are construction worksites, but not all programs are successful. What makes a successful safety and health program? There are seven elements:

1. **Commitment**: All employees- including company executive officers, managers, and supervisors- are committed to making the program succeed.

2. **Accountability**: All employees - including company executive officers, managers, and supervisors - are held accountable for following safe work practices.

3. **Involvement**: All employees, including managers and supervisors, participate in making the program succeed.

4. **Hazard identification**: All employees are trained to identify hazards, and there are procedures for conducting hazard inspections and reporting hazards.

5. **Accident investigation**: Managers and supervisors promptly investigate all accidents and near misses, and then determine how to eliminate their causes.

6. **Training**: All employees receive training in identifying construction worksite hazards and learning safe work practices.

7. **Evaluation**: Managers and supervisors, with help from other employees, evaluate the program's strengths and weaknesses at least once a year.

1. Which fall protection program element is most likely to be closely inspected and cited by OSHA?

a. Hazard identification
b. Accountability
c. Training
d. Involvement

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Prepare a Safety and Health Policy

Does your company have a written safety and health policy? It should. A written policy does the following:

- it reflects a commitment to a safe and healthful worksite,
- it summarizes management and employee responsibilities, and
- it emphasizes the safety and health program's role in achieving that goal.

Keep the policy brief, commit to it, and enforce it. Look at a sample policy.

Competent Persons (CP) and Qualified Persons (QP)

Certain activities throughout OSHA's standards require completion by either a Competent Person (CP) or a Qualified Person (QP). These persons must have the training and expertise to evaluate hazardous conditions, inspect equipment, evaluate mechanical systems, or train others to work safely.

Competent Person

A competent person is one who:

- is capable of identifying existing and predictable hazards in the surroundings or working conditions that are unsanitary, hazardous, or dangerous to employees; and
- has authorization to take prompt corrective measures to eliminate them. [1926.32(f)]

A competent person must have authority to take prompt measures to eliminate hazards at the work site and have the experience to be capable of identifying those hazards.

Instruction, training, and experience are ways that an employee can become competent, but attending lectures isn't enough. The employee also needs to be able to demonstrate their capabilities to the employer.

Qualified Person

A qualified person is one who:

- possesses a recognized degree, certificate, or professional standing, or has extensive knowledge, training, and experience, and

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- has successfully demonstrated the ability to solve or resolve problems relating to the subject matter, the work, or the project. [1926.32(m)]

A qualified person must have a recognized degree, profession certificate, or extensive experience and ability to solve the subject problems at the worksite. There may be a requirement for more technical or engineering knowledge.

For example, a competent person can identify electrical components and have a general knowledge of electrical hazards. However, a qualified person would have additional knowledge, skills, and training to do specialized electrical work tasks.

2. As defined by OSHA, a "qualified person" must possess _____.

a. 10-hours of training in specific safety procedures
b. an OSHA 30-Hour Outreach Course certificate
c. at least five years of on-the-job experience
d. a recognized degree, certificate, or extensive experience

Choosing a Competent or Qualified Person

Although Federal OSHA defines competent and qualified persons, it does not provide specifics for determining who can assume these roles.

To help determine if someone meets the qualifications as a competent or qualified person, the employer can do the following:

- Know the OSHA rules that apply to your construction worksite. The rules will tell you if you need to designate a competent or a qualified person.

- If an OSHA rule that applies to your construction worksite requires a competent or a qualified person, note duties and responsibilities that the rule requires the person to perform.

- If an OSHA rule that applies to your construction worksite requires a competent person, that person must have the authority to take prompt corrective measures to eliminate hazards.

- Determine the knowledge, training, and experience the competent or qualified person needs to meet the rule's requirements.

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- Designate a person who has the knowledge, training, and experience that meets the rule's requirements.

3. Before employers designate competent and qualified persons, they should determine the required _____.

e. duration of the project
f. knowledge, skills, and abilities
g. OSHA schedule of inspections
h. number of employees on the worksite

Duties and Responsibilities

The Competent Person

- Serves as the monitor in a safety-monitoring system and is responsible for recognizing hazards that cause falls and warning workers about them.
- Determines that safety nets meet minimum requirements.
- Inspects a personal fall-arrest system after it arrests a fall and determines if the system is undamaged and can be used again.
- Evaluates any alteration in a personal fall-arrest system and determines if the system is safe to use.
- Supervises the installation of slide-guard systems.
- Trains employees to recognize hazards that cause falls and to follow procedures that minimize the hazards.

The Qualified Person

- Supervises design, installation, and use of horizontal lifeline systems to ensure that they can maintain a safety factor of at least two - twice the impact of a worker free-falling 6 feet.
- Supervises design, installation, and use of personal fall-restraint anchorages.
- Supervises design, installation, and use of personal fall-arrest anchorages.

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4. Which of the following responsibilities is assigned to a qualified person?

a. Supervise the design, installation, and use of anchorages
b. Supervise the installation of slide-guard systems
c. Determine that safety nets meet minimum requirements
d. Determine if fall-arrest systems are undamaged and suitable for reuse

Scenario

On September 5, 2006, a 30-year-old construction laborer (the victim) was fatally injured when he fell through a floor opening to a concrete floor approximately 10 feet, 10 inches below.

The victim had spent the majority of his work shift cutting plywood sheathing and handing it up to coworkers who were sheathing the roof. He was working from the second floor of a two-story, single-family home under construction where he and other workers had completed most, but not all, of the subfloor (plywood sheathing secured over floor joists) on previous workdays.

This photograph shows the two-story home workers were framing when the accident occurred.

Photo: Courtesy: NCDOL/OSH

They had left a floor area open in an attic space where a walk-in closet was to be constructed later. At the end of the shift, the victim's lead worker asked two workers to complete the second story subfloor in the attic space. One of these workers joined the roofers to lay roofer's felt. The other co-worker joined the victim and together, they cut two sheets of plywood sheathing and placed them over the joists in the open area in the attic space.

They were not wearing fall protection. The co-worker reported he was looking down at the sheets of unsecured plywood sheathing trying to make the pieces fit into an opening that was not square and when he looked up the victim was gone. The victim had apparently stepped onto a piece of the unsecured plywood sheathing that covered part of the floor opening, and when the plywood sheathing pivoted on the floor joist, he fell through the opening.

Conclusion

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- Ensure all employees are provided with and use appropriate fall protection when exposed to fall hazards.
- Ensure through employee training and job-site inspection that correct construction procedures, such as use of appropriate fasteners, are followed during all phases of construction.
- Develop, implement, and enforce a comprehensive, written fall protection program that, at a minimum, complies with applicable OSHA fall prevention standards.
- Assign a competent person to inspect the worksite before work begins to identify fall hazards and to determine the appropriate fall prevention systems for workers.

5. When should the competent person inspect the worksite for fall hazards?

a. At least daily as needed
b. After each workday is finished
c. Each morning before work begins
d. Before and after the workday

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Module 3: Fall Hazard Risk Analysis

Analyze the Risk of Falling

To effectively prevent and control fall hazards on the construction site, you need to conduct a 4-step risk analysis process. Using the risk analysis process, do the following:

1. identify the actual and potential fall hazards on the construction project;
2. determine the unique characteristics of each hazard;
3. evaluate the danger each hazard poses to workers; and
4. prioritize or rank each hazard according to its overall risk.

Identifying Hazardous Work Areas

It's important to identify areas in which workers' tasks could expose them to fall hazards.

Ask if workers will be using portable ladders, supported scaffolds, aerial lifts, or suspension platforms to reach their work areas. You will also want to know how and where workers will use the equipment. To answer these questions:

- meet during the preplanning phase to determine the construction activities that will be occurring on the worksite; and
- after construction starts, and in every phase after that, get on-site to identify the current and potential hazards that could exist in future phases.

Look at each phase of the project from the ground up. Ensure that all walking/working surfaces have the strength to support workers and their equipment and then identify all tasks that could expose workers to falls. A walking/working surface is any surface, horizontal or vertical, where a person walks or works.

Hazardous Areas

Here are some examples of areas on a worksite that have a track record of causing injuries:

- holes in walking/working surfaces that they could step into or fall through
- elevated walking/working surfaces six feet while performing construction and four feet for general industry (non-construction) or more above a lower level

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- skylights and smoke domes that workers could step into or fall through
- wall openings such as those for windows or doors that workers could fall through
- trenches and other excavations that workers could fall into
- walking/working surfaces from which workers could fall onto dangerous equipment
- hoist areas where guardrails have been removed to receive materials
- sides and edges of walking/working surfaces such as established floors, mezzanines, balconies, and walkways that are 6 feet or more above a lower level and not protected by guardrails at least 39 inches high
- ramps and runways that are not protected by guardrails at least 39 inches high
- leading edges - edges of floors, roofs, and decks - that change location as additional sections are added
- wells, pits, or shafts not protected with guardrails, fences, barricades, or covers

1. When should the employer meet with contractors to determine expected construction activities?

a. When the construction contract is signed
b. Prior to any anticipated OSHA inspection
c. When work begins on the project
d. During the project preplanning phase

Identifying Fall Hazards

A fall hazard is anything in the workplace that could cause an accidental loss of balance or bodily support that results in a fall.

Fall hazards cause accidents such as the following:

- A worker walking near an unprotected leading edge trips over a protruding board.
- A worker slips while climbing an icy stairway.

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- A makeshift scaffold collapses under the weight of four workers and their equipment.
- A worker is carrying a sheet of plywood on a flat roof and steps into a skylight opening.

Fall hazards are foreseeable. You can identify and control them before they cause injuries. The hazards that caused the above accidents could have been eliminated or controlled had they first been identified, evaluated, and corrected.

What are possible solutions that would prevent the falls described above?

Solutions

1. Worker #1 was either distracted or did not recognize and correct the hazards. Had he been aware, he could have taken proper fall protection measures.
2. Worker #2 did not identify and evaluate the hazard. He would have then taken measures to prevent slipping.
3. Had the scaffold in #3 been properly inspected and evaluated before the workers and equipment loading it, the workers would not have been allowed to work on the scaffold.
4. If the roof in #4 had been inspected and holes properly guarded, the fall through the skylight would not have occurred.

None of the hazards in the examples given above would have caused an injury if someone had identified, evaluated, and controlled them.

2. None of the hazards on a construction worksite can cause an injury if they are first _____.

a. reported to OSHA for inspection
b. identified and noted on an inspection report
c. identified, evaluated, and corrected
d. evaluated by the worksite supervisor

Analyzing Fall Hazards

After identifying an existing or potential fall hazard, it's important to analyze the hazard by looking at its unique characteristics. Two important fall-hazard characteristics to analyze to determine risk are probability and severity.

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Determine Probability

1. The probability, or likelihood, that a hazard will cause a fall.

2. Distance of the fall. A person in free-fall will accelerate until he or she reaches a terminal velocity of about 120 mph.

3. Speed of deceleration. The faster you stop or decelerate at impact, the greater the impact forces on the body and severity of injury.

4. Nature of the surface. The "hardness" of the surface upon which the worker falls affects the intensity of the impact and the severity of injury.

5. Orientation of the body at impact. The severity of the injury also depends on the position of the body when it strikes the surface, and that is a matter of luck more than anything else.

Determine Severity

An important part of hazard analysis is to determine how severe an injury might be as a result exposure to a hazard. Estimating the severity of an injury after a fall is difficult because it is depending on many variables. A basic rule regarding the severity of an injury after a fall is that "it's not the fall that gets you, it's that sudden deceleration on impact."

1. **Distance of the fall.** A person in free-fall will accelerate until he or she reaches a terminal velocity of about 120 mph.

2. **Speed of deceleration.** The faster you stop or decelerate at impact, the greater the impact forces on the body and severity of injury.

3. **Nature of the surface.** The "hardness" of the surface upon which the worker falls affects the intensity of the impact and the severity of injury.

4. **Orientation of the body at impact.** The severity of the injury also depends on the position of the body when it strikes the surface, and that is a matter of luck more than anything else.

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3. Which two basic characteristics of a fall hazard do you analyze to determine risk?

a. The severity of injury and duration of exposure
b. The probability and severity of injury
c. The number of workers and probability
d. The orientation of the body and fall distance

Evaluating Fall Hazards

Determine Fall Distances

Part of the hazard analysis process is to determine fall distances from walking-working surfaces to lower levels so that the employer can provide appropriate fall protection equipment. Let's look at the various fall-distance rules.

Four-Foot Rule: OSHA general industry standard 1910.28(b), Walking-Working Surfaces, requires workers be protected from fall hazards on walking-working surfaces where they could fall four feet or more to a lower level.

When working less than 4 feet above dangerous equipment, guardrail systems, travel restraint systems, or equipment covers must be used. When working 4 feet or higher above dangerous equipment, guardrail systems, travel-restraint systems, fall-arrest systems, safety nets, or must be used.

Six-Foot Rule: OSHA's standard 1926.501(b) requires the use of fall protection when construction workers are working at heights of 6 feet or greater above a lower level. However, the rule does not apply to workers inspecting, investigating, or assessing workplace conditions prior to the actual start of work or after all construction work has been completed.

When working six feet or less over dangerous equipment, guardrail systems of equipment guards must be used. When working six feet or higher over dangerous equipment, workers must be protected using guardrail systems, personal fall arrest systems, or safety nets.

Below are some examples of fall hazards from which employees must be protected by the four-foot and six-foot rules.

- Holes and skylights in walking-working surfaces
- Wall openings that have an inside bottom edge less than 39 inches above a walking-working surface

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- Established floors, mezzanines, balconies, and walkways with unprotected sides and edges

The four-foot rule does not apply:

- to work on portable ladders;
- when inspecting, investigating, or assessing work condition before or after work;
- entertainment stages or rail-station perimeters;
- powered platforms;
- telecommunications work; and
- electric power generation, transmission, and distribution.

Ten-Foot Rule: According to 1926.451(g)(1), fall protection on scaffolding is required when you are greater than 10 feet off the lower level.

Fifteen-Foot Rule: According to 1926.760, Steel Erection, each employee on a walking/working surface with an unprotected side or edge more than 15 feet above a lower level must be protected from fall hazards by guardrail systems, safety net systems, personal fall arrest systems, positioning device systems or fall restraint systems.

4. The "six-foot rule" for fall protection applies to _____.

a. mining operations
b. agriculture
c. general industry
d. the construction industry

Controlling Exposure and Behaviors

It's important to understand that exposure to fall hazards can be eliminated, prevented, or controlled, especially when a combination of methods are used.

Controlling exposure to hazards in the workplace, no matter what the situation, is generally done using a strategy called the "Hierarchy of Controls" (HOC). We have developed a modified fall prevention HOC model that combines the recommendations in ANSI/ASSP Z359.2, Minimum Requirements for a Comprehensive Managed Fall Protection Program, ANSI/ASSP Z10, Occupational Health and Safety Management Systems, and OSHA's Hierarchy of Controls.

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Exposure Controls

Exposure controls focus on strategies and methods to prevent or control exposure to fall hazards. If used alone, exposure controls are not as reliable because workers don't know why they are important.

1. **Elimination/Prevention**: For example, you can eliminate exposure to a hazard by lowering the work surface to ground level, or move a process, sequence or procedure to a different location so that workers no longer need to approach a fall hazard. Elimination is most effective because if you can remove the hazard, you also eliminate the exposure.

2. **Passive fall protection**: Barriers such as guardrails, covers over exposed floor openings, and safety nets are used to isolate or separate the fall hazard from the worker.

3. **Active fall protection**: These controls actively prevent access to fall or, if a fall occurs, limit the distance of the fall.

    - **Travel restraint systems**: Positioning and restraint equipment such as full-body harnesses and lanyards tied to anchors of lifelines can keep the worker away from a ledge or other fall hazard.

    - **Fall arrest systems**: Fall protection equipment such as fall-arrest systems and safety nets are used to prevent injury should a fall occur.

Behavioral Controls

Behavioral controls focus on controlling worker behaviors to prevent falls. If used alone, behavioral controls are not as reliable as exposure controls.

4. **Warnings:** Warnings are used to raise awareness of fall hazards. Warning tape and cones are common types of warning.

5. **Administrative controls:** These methods do not focus on controlling exposure to hazards, but on controlling workers behaviors. Administrative controls are last on the hierarchy of controls because they rely on worker compliance for success. And as we know, any system that relies on human behavior is inherently unreliable.

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5. Which of the following would totally eliminate the risk of a fall to a lower work surface?

a. Place guard rails around the elevated surface
b. Lower the work surface to ground level
c. Substitute fall arrest with fall restraint equipment
d. Enforce safety rules at all times

Exposure Controls

Elimination/Prevention

Elimination is more difficult, but it is by definition the most effective. Complete elimination of a fall hazard would mean not putting any workers at heights. Remember, no hazard, no harm. You can also prevent exposure to fall hazards by designing the job using methods that reduce the probability or severity of a fall. The best time to design the prevention of fall hazards on the construction project is in the preplanning phase when it is less expensive to implement.

Passive Fall Protection

Passive control methods are considered a higher level of protection from falls than active control methods because, when installed, they do not rely on the actions of a worker to be successful.

Active Fall Protection

Active control methods are considered to be less effective because they rely on the worker to use equipment to be successful.

Elimination

- assembling sections of a scaffold on the ground before being lifted into place by a crane;
- using tools and devices with extended handles from ground-level to cut, grab, or move items at elevation; and
- using long extension poles to spray paint or change ceiling light bulbs.

Prevention Through Design

- building stairway systems to eliminate the need to climb ladders;
- using articulating and scissor lifts to replace the need to use fall-arrest systems;

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- using equipment that reduces the probability of exposure to a fall hazard; and
- moving a process, sequence or procedure to a different location so that workers no longer need to approach a fall hazard.

Passive Controls

Guardrails: Guardrails are the most common fall protection system used to protect workers at openings and on platforms. Guardrails must be continuous, with no portion of the leading edge exposed, and meet the strength requirements. They should be removed only when materials are being on-loaded or off-loaded. Ensure that each top rail, or equivalent member of a guardrail system, can withstand a force of at least 200 pounds. Top rails must be between 39 and 45 inches in height. Midrails, screens, mesh, intermediate vertical members or panels must withstand a force of at least 150 pounds.

Hole Covers: Any hole opening that measures at least two inches or more in its least dimension in a floor, roof or walking surface must be covered. Covers should be able to support at least two times the weight that will cross over them.

Safety Net Systems: Safety nets can catch the worker after a fall. They are most often found under bridgework, steel erection, demolition and maintenance operations. They should be installed as close as possible under the walking/working surface, but never allow a fall of more than 30 feet.

Active Controls

Personal Fall Restraint Systems: This active fall system can be a full body harness or a body belt that's rigged to physically restrict the worker from reaching an area where a fall could occur. It's designed to hold the worker back, but it does not provide support if a fall occurs. Fall restraint systems are commonly used on leading edge work such as roofs, open-sided floors and work platforms.

Personal Fall Arrest Systems (PFAS): A PFAS keeps the worker from hitting the ground or other obstructions below. The PFAS consists of a body harness, connecting device and anchorage. It must prevent the worker from free falling more than six feet. The arresting force on the worker must be limited to 1,800 pounds or less, and the anchorage point must be capable of supporting 5,000 pounds per person attached. When vertical lifelines are used, each worker must be attached to a separate lifeline.

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6. Passive control methods are considered a higher level of protection from falls than active control methods because _____.

a. they are less expensive to use
b. they may be used multiple times
c. workers are required to use the equipment
d. they do not rely on the actions of workers

Behavioral Controls

Warnings

Warnings are usually audible or visible devices that help to raise awareness of hazards. Signs, labels, posters, and lights are examples of warnings that alert workers about hazards. The effectiveness of warnings is highly dependent on the quality of training, legibility and visibility, and worker compliance. Warnings may become ineffective if, over time, workers ignore them.

Administrative Controls

Administrative controls are changes made to the way employees work. In construction, administrative controls are the policies, programs, plans, processes, procedures, and practices aimed at reducing worker exposure to hazards that elimination, passive fall controls, and active fall controls fail to eliminate. Effective administrative controls have the potential to successfully eliminate the behaviors that result in over 90% of all workplace accidents!

It's always better to eliminate the hazard so that you don't have to rely on administrative controls because they are only effective when workers comply. It's important to understand that controls that rely on human behavior are inherently less reliable.

Administrative controls to help ensure safe behaviors include safety:

- instruction, training, and practice to promote workers have adequate knowledge, skills, and abilities (KSAs) to work safely;
- policies that promote decision-making at the supervisor and employee levels, (e.g., job scheduling to limit exposure);
- processes that help to ensure complex operations are safe (e.g., chemical production preventive and corrective maintenance);
- procedures that ensure that the steps in single operation are safe (e.g., performing lockout/tagout and housekeeping);

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- practices ensure behaviors in each step in a procedure are safe (e.g., inspecting PPE after use);
- rules that ensure mandatory safety behaviors (e.g., requiring the use of PPE when handling hazardous chemicals); and
- guidelines, to promote non-mandatory safety behaviors (e.g., submitting safety suggestions).

Safe work practices include:

- removing tripping, blocking, and slipping hazards
- removing accumulated toxic dust on surfaces
- wetting down surfaces to keep toxic dust out of the air
- using safe lifting techniques
- maintaining equipment and tools in good repair
- using personal protective equipment (PPE)

To make sure administrative controls are effective in the long term, they must be designed and used in conjunction with, and not as a substitute for, more effective or reliable hazard controls.

7. Administrative controls are not as effective as elimination, passive controls, or active controls because they _____.

a. can't be used as a basis for discipline
b. make the employer liable for all accidents
c. work only as long as workers comply
d. are expensive and not referenced often

Scenario

A project engineer was fatally injured by a fall 29 feet from a roof while measuring the roof for an insulation cost estimate. The victim fell over the edge of the root while walking backwards. A maintenance person from the building who had accompanied the victim and two co-workers to the roof immediately ran downstairs and called 911 from the office and

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proceeded to the victim to offer assistance. Emergency medical services arrived immediately. The victim was transported to a nearby local hospital and was pronounced dead on arrival.

Employers should:

- employ alternative controls for fall hazards when personal fall-arrest systems are not required nor appropriate
- develop, implement, and enforce a comprehensive safety program that includes, but is not limited to, training all employees in fall hazard recognition

Building owners should:

- consider the installation of guardrails at the perimeter of flat roofs wherever possible

8. When fall-arrest systems are not required nor appropriate, what should the employer do when employees are exposed to fall hazards?

a. Remind the employee to use common sense
b. Use alternative controls
c. Call OSHA to get permission
d. Tell the worker to be safe

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Module 4: Supported and Suspended Access

Supported Access

Ladders, supported scaffolds, and aerial lifts let you get to a work area and support you while you work. They make getting to a work area easy, but they can cause falls when they are not used properly.

Portable Ladders

Portable ladders are versatile, economical, and easy to use. However, workers sometimes use them without thinking about using them safely. Each year, most workers are injured when they fall from ladders. Most of the falls are less than 10 feet.

Types of portable ladders: We use ladders to do all sorts of tasks, so it's not surprising that many types of ladders are available. Let's look at the most common types.

Straight Ladder (left)

The most common type of portable ladder. Length cannot exceed 30 feet. Available in wood, metal, and reinforced fiberglass. Supports only one worker.

Standard Folding Ladder (right)

Folding ladders have flat steps, a hinged back, and is not adjustable. For use only on firm, level surfaces. Available in metal, wood, or reinforced fiberglass. Must have a metal spreader or locking arm and cannot exceed 20 feet. Supports only one worker.

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Extension Ladder (left)

Extension ladders offer the most length in a general-purpose ladder. They have two or more adjustable sections. The sliding upper section must be on top of the lower section. Made of wood, metal, or fiberglass. Maximum length depends on material. Supports only one worker.

Platform Ladder (right)

Platform ladders have a large, stable platform near the top that supports one worker. Length cannot exceed 20 feet.

Trestle Ladder (left)

Trestle ladders have two sections that are hinged at the top and form equal angles with the base. Used in pairs to support planks or staging. Rungs are not used as steps. Length cannot exceed 20 feet.

Tripod (Orchard) Ladder (right)

Tripod ladders have a flared base and a single back leg that provides support on soft, uneven ground. Length cannot exceed 16 feet. Metal and reinforced fiberglass versions are available. Supports only one worker.

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1. Which type of ladder cannot exceed 30 feet?

a. Folding ladder
b. Trestle ladder
c. Straight ladder
d. Extension ladder

How Ladder Falls Occur

Most workers fall from unstable ladders that shift or tilt when the workers climb too high or reach too far beyond the side rails. Workers also fall when they slip on rungs while they're climbing or descending and when vehicles strike the ladders.

Keep the following in mind to stay safe when you use a portable ladder:

- select the most appropriate ladder for the task;
- inspect the ladder before using it; make sure it's in good condition;
- angle straight ladders and extension ladders properly with a 1:4 slope (base to height);
- protect the base of a ladder to prevent others from accidentally striking it;
- Select a ladder that will extend at least 3 feet (36 inches) above the landing point;
- If the ladder could be displaced by work activities, secure it;
- face the ladder when you climb or descend it using the three-point-control method;
- stay within the side rails when climbing or working from the ladder;
- if you do reach out to the side, keep the rest of your body within the rails;
- raise and lower heavy loads with a hand line or a hoist;
- make sure metal ladders have steps and rungs with skid-resistant surfaces;
- allow only one person on the ladder;
- never stand on to top rung of the ladder;

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- never use ladders that have conductive side rails near exposed energized equipment; and
- never use ladders on scaffolds to extend reach.

Required Ladder Safety Training

As with any safety training in the workplace, safe ladder use training should ensure construction workers have adequate knowledge, skills, and abilities to safely use portable ladders.

Before workers use ladders, a competent person must train them so that they understand all of the best practices we mentioned above and the following:

- the nature of the fall hazards in the work area;
- how to use, place, and care for ladders; and
- maximum intended load-carrying capacities of the ladders.

2. If the upper landing point is 12 feet high, how many feet should you place the ladder away from the landing point?

a. 1 foot
b. 2 feet
c. 3 feet
d. 4 feet

Supported Scaffolds

Supported scaffolds consist of one or more platforms supported by outrigger beams, brackets, poles, legs, uprights, posts, frames, or similar rigid support.

If you work on a supported scaffold more than 10 feet above a lower level, you must be protected from falling.

Guardrails are appropriate for most scaffold platforms. If you can't use a guardrail system, then you must use personal fall-arrest systems or restraint systems.

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Supported Scaffold Types

Of the many types of supported scaffolds, fabricated/frame scaffolds are the most common. Like portable ladders, they're versatile, economical, and easy to use. You'll see them on construction sites as single supported platforms and multiple platforms stacked several stories high on modular frames.

Frame Scaffold or Fabricated Frame. Platform(s) supported on fabricated end frames with integral posts, horizontal bearers, and intermediate members.

Tube/Coupler. Platform(s) supported by tubing, erected with coupling devices connecting uprights, braces, bearers, and runners.

Mast Climber. Mast climbing supported scaffolds (Mast Climbers) carry much heavier loads than traditional scaffolding and serve to position personnel, along with necessary tools, equipment, and materials to perform work at great heights. Click to enlarge image. Three construction employees were killed when the left mast's platform on the left side collapsed at a project under construction.

Ladder Jack Scaffold. A ladder jack scaffold is a simple device consisting of a platform resting on brackets attached to a ladder. Ladder jacks are primarily used on lighter loads because of their portability and cost-effectiveness.

Mobile (Manually or Propelled). Unpowered, portable, caster- or wheel-mounted supported scaffold.

Specialty Scaffold. Scaffold types designed for a narrow and very specific range of applications. Other specialty scaffolds include:

- Plasterers', Decorators', and Large-Area Scaffolds
- Bricklayers' Square Scaffolds
- Horse Scaffolds
- Form Scaffolds and Carpenters' Bracket Scaffolds
- Outrigger Scaffolds
- Window Jack Scaffolds
- Crawling Boards (Chicken Ladders)
- Step, Platform, and Trestle Ladder Scaffolds

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- Stilts

Pump Jack Scaffold. Platform supported by vertical poles and movable support brackets.

3. At what height on a supported scaffold must fall protection be provided?

a. Over 4 feet from the ground
b. More than 10 feet above a lower level
c. Over 6 feet from a lower work surface
d. Any level above 2 feet from the ground

How Suspended Scaffold Falls Occur

Workers fall from suspended scaffolds when components fail, planks break, handrails give way, and scaffold supports collapse. The Bureau of Labor Statistics' Census of Fatal Occupational Injuries (CFOI) reported 564 fatal falls to a lower level in 2019. Among those were falls from scaffolds. However, most scaffold accidents can be ultimately traced to a lack of knowledge, skills, and abilities (KSAs) because workers are untrained or improperly trained.

Scaffold workers must be trained on the requirements and safe practices to recognize fall hazards and to control or minimize the hazards.

Required topics for supported scaffold safety training include:

- scaffold load capacity and the types of loads appropriate for the scaffold;
- when fall protection is required, the appropriate protection to use, and how to use it;
- how to use scaffold components;
- how to reach access areas;
- how to protect those below the scaffold from falling objects; and
- how to avoid electrical hazards.

Effective supported scaffold safe work practices include:

- Use ladders or stairs to reach platforms that are more than 2 feet above or below the access point.
- Don't climb cross-braces to reach a scaffold platform.

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- Scaffolds must be able to support their own weight and at least four times the maximum intended load. The maximum intended load includes workers, equipment, and supplies.
- Platforms must not deflect more than 1/60 of the span when they are loaded.
- Platforms must be fully decked or planked between the front uprights and the guardrail supports.
- Don't use damaged scaffold components; repair or replace them immediately.
- Make sure a competent person inspects the components before each workshift.
- Don't modify components.
- Scaffold components made by different manufacturers may be mixed, provided they fit together without force and maintain structural integrity.
- Watch for slippery surfaces. Don't work on platforms covered with snow and ice.
- Stay off scaffolds during storms and strong winds unless a competent person determines that it's safe.
- Keep a safe distance from power lines and any other conductive source.

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Minimum clearance distances:

Insulated lines		
Some overhead power lines are installed with a covering that insulates the voltages in the line. The insulation usually deteriorates over time.		
Voltage	**Minimum distance**	**Alternatives**
Less than 300 volts	3 feet	–
300 volts to 50 kv	10 feet	–
More than 50 kv	10 feet plus 0.4 inches for each 1 kv over 50 kv	Two times the length of the line insulator, but never less than 10 feet
Uninsulated lines		
Voltage	**Minimum distance**	**Alternatives**
Less than 50 kv	10 feet	–
More than 50 kv	10 feet plus 0.4 inches for each 1 kv over 50 kv	Two times the length of the line insulator, but never less than 10 feet

- Scaffolds must be erected, dismantled, or moved only under the supervision of a competent person.
- The competent person must be on site to direct and supervise the work.

4. Which of the following topics is required in supported scaffold safety training?

a. Scaffold load capacity
b. Scaffold enforcement policies
c. Scaffold design advantages
d. Scaffold job hazard analysis

Mobile Elevating Work Platforms (MEWPs)

"Mobile Elevating Work Platform" (MEWP) is a term, adopted by ANSI in 2015, to describe various aerial work platforms. MEWPs are mechanical devices used to provide temporary access for people or equipment to inaccessible areas, usually at height. MEWPs are also known as aerial lift devices, elevating work platforms, or bucket trucks.

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Types of Devices and Platforms

Most devices and platforms have extensible or articulating mechanisms that can position workers up, down, or sideways. The safe use and design of the types of elevating devices and platforms are covered by ANSI standards.

MEWP Types

MEWP Type is determined by whether the lift can travel when stowed or elevated and the location of the controls which allow such travel.

- Type 1 MEWPs can only travel with the platform in a stowed position
- Type 2 MEWPs can travel elevated and is controlled from the chassis
- Type 3 MEWPs can travel elevated and is controlled from the platform
- Group A - vertical projection of the platform is inside the tipping lines at maximum inclination in all platform configurations
- Group B - all other aerial lifts, typically identified as boom-type aerial lifts

The types of elevating devices and platforms covered by ANSI standards include:

- vehicle-mounted elevating and rotating Aerial Devices (ANSI A92.2 devices);
- manually propelled elevating aerial platforms (ANSI A92.3 devices);
- boom-supported elevating work platforms (ANSI A92.5 devices);
- self-propelled elevating work platforms and scissor type MEWPs (ANSI A92.6 devices);
- airline Ground Support Vehicle-Mounted Vertical Lift Devices (ANSI/SAIA A92.7);
- vehicle-Mounted Bridge Inspection and Maintenance Devices (ANSI/SAIA A92.8);
- mast-Climbing Work Platforms (ANSI/SAIA A92.9);
- transport Platforms (ANSI/SAIA A92.10); and
- safe Use of Mobile Elevating Work Platforms (MEWPs) (ANSI/SAIA A92.22).

For more information on the various types of MEWPs, visit the MEWPs Inc. website.

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How MEWP Falls Occur

The root cause for most accidents involving MEWPs can be traced to untrained or improperly trained workers who lack the knowledge, skills, and abilities (KSA) to safely use the equipment.

The causes of falls from MEWPs include:

- hydraulic cylinder fails and causes the boom to drop;
- outriggers are not used or improperly placed and the lift vehicle overturns;
- workers are not tied off while they are in the bucket; and
- workers fall or are pulled off the platform when the lift vehicle is struck by another vehicle or moves unexpectedly.

5. What is the root cause for most Mobile Elevating Work Platform (MEWP) accidents?

a. Scaffold component failure
b. Inadequate knowledge, skills, and abilities (KSAs)
c. Inappropriate worker behaviors
d. Poor employer enforcement policies

MEWP Fall Protection

If you work from a MEWP, you must be protected from falling. The type of fall protection you need depends on the type of MEWP you use. Most platforms must have a guardrail and each worker may be required to use a personal fall-arrest system. Other considerations include the following:

- When operating boom lifts or aerial bucket trucks a full body harness and shock absorbing lanyard should be used.
- For scissor-lifts and other vertical lift equipment, a body-belt with a non-shock absorbing lanyard is acceptable.
- A rescue plan should be developed in case the MEWP does not operate properly or a worker may fall requiring rescue.

MEWP Safe Practices

The workers in the image above might be seconds away from a serious or fatal accident because they are not using equipment as designed, not using required fall protection, nor

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following safe work practices. Would a boom-supported elevating work platform be more appropriate for this job?

Keep in mind the following when you use a MEWP:

- Use the MEWP only for its intended purpose and follow the manufacturer's instructions. Keep the operating manual with the MEWP.
- Keep the MEWP level and stable; use outriggers and intermediate stabilizers.
- Never move the MEWP when the boom is up and workers are on the platform.
- Stand on the platform floor. Don't sit or climb on the edge of the basket, guardrail, or midrail.
- Be sure to close the access gate while you're working from the platform.
- Inspect the MEWP before using it to make sure that it's working properly and is in good condition.
- Know the MEWP's rated load capacity and don't exceed it.
- Stay at least 10 feet away from electrical power lines.
- Never use the MEWP during severe weather.
- Use warning signs or barricades to keep others out of the work area.
- Never tie off to equipment or to a structure next to the platform.

For more information on the safe use of MEWPs, see ANSI/SAIA A92.22, Safe Use of Mobile Elevating Work Platforms (MWEPs).

6. On which type of mobile elevating work platform (MEWP) is a body-belt with a non-shock absorbing lanyard acceptable fall protection?

a. Cherry pickers
b. Aerial bucket trucks
c. Boom lifts
d. Scissor-lifts

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Suspended Access

When supported access to a work area is not feasible or safe, an alternative is suspended access. Typically, on construction sites, suspended scaffolds are used to gain access. A suspended scaffold is a temporary elevated platform that hangs by wire rope, or other non-rigid means, from an overhead structure.

Suspended Scaffold Types

Two-point scaffolds are the most common type of suspended scaffold, and the fall protection requirements for these scaffolds also apply to most other types. Requirements specific to the other types are described only in their respective modules.

Catenary. Platform supported by two essentially horizontal and parallel ropes attached to structural members of a building. Additional vertical pickups may also provide support.

Float (Ship). Braced platform resting on two parallel bearers and hung from overhead supports by ropes of fixed length.

Interior Hung. Platform suspended from the ceiling or roof structure by fixed-length supports.

Multi-level. Posts with fixed connection points that accept runners, bearers, and diagonals, also made of wood, that can be interconnected at predetermined levels.

Multi-point Adjustable. Platform(s) suspended by more than two ropes from overhead supports and equipped with a means to permit the platform to be raised and lowered. Includes chimney hoists.

Needle Beam. A platform suspended from needle beams.

Single-point Adjustable. Platform suspended by one rope from an overhead support and equipped with a means to permit the platform to be moved to desired working levels.

Two-point (Swing Stage). Platform supported by hangers (stirrups) suspended by two ropes from overhead supports and equipped with a means to permit the platform to be raised and lowered.

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7. Typically, why are suspended scaffolds used on construction sites?

a. To work over uneven surfaces
b. To work at various elevations
c. To gain access to work areas
d. To allow multiple workers a platform

Adjustable-Suspension Scaffolds

Add a hoist to move the platform up or down, and you have an adjustable-suspension scaffold - but not necessarily a safe one. Suspension ropes, lifelines, platforms, hoists, overhead support devices, and tieback systems are critical to the safety of adjustable-suspension scaffolds.

Before you use an adjustable-suspension scaffold, you need to know how to rig the scaffold, how to operate the hoist, how to work safely from the scaffold, and what to do in an emergency. Other requirements include the following:

- A competent person must examine all direct connections that are part of the system and confirm that the connections will support the platform loads.
- You must also wear a personal fall-arrest system to protect yourself if a connection fails.

Most accidents involving adjustable-suspension scaffolds happen when a primary suspension rope breaks and workers are not wearing personal fall-arrest gear. Steel suspension ropes rarely break if they're correctly rigged, maintained, and inspected regularly. If you work on an adjustable-suspension scaffold more than 10 feet above a lower level, you must be protected from falling.

Causes of Falls

- ascending platforms snag and electric hoists continue to operate snapping weak ropes;
- pressure from the two steel discs that clamp support ropes in sheave-type hoist motors can break a weak rope;
- failing anchors also cause serious accidents;
- workers don't use personal fall-arrest systems as required;
- untrained workers attach lifelines and suspension ropes to unapproved rooftop fixtures; and

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- lifelines fail because they are hung over unpadded edges.

Protection from Falls

- **Single-point and two-point adjustable-suspension scaffolds:** Personal fall-arrest systems and guardrail systems are required on single-point or two-point adjustable-suspension scaffolds. The top edge of the guardrail must be between 36 inches and 45 inches above the platform surface. (The top edge can exceed 45 inches when necessary.)
- **Multipoint adjustable-suspension scaffolds:** Personal fall-arrest systems and guardrail systems are required on multipoint adjustable-suspension scaffolds. The top edge of the guardrail must be between 36 inches and 45 inches above the platform surface. (The top edge can exceed 45 inches, when necessary.)

8. When working on an adjustable-suspension scaffold, at what height must you be protected from falling?

a. Six feet or higher above a work surface
b. More than 10 feet above a lower level
c. Four feet above ground level
d. At any height when suspended

Descent-Control Devices

A descent-control device lets you descend a primary support rope - typically from a boatswain's chair - then lock the device when you reach the work area. The device works by friction, engaging the support rope and controlling the descent speed.

Most workers start from the roof and work down the face of the building. When they reach the ground, they remove the descent equipment from the support rope and return to the roof for another drop. Click the image to the right to see the descent device more clearly.

Most falls result from failure of the primary support rope or a supporting anchor, not the descent device. Support ropes fail because workers don't inspect them regularly or they misuse them. Anchors fail when workers simply assume, they are secure. Descent devices, support ropes, and anchors rarely fail when workers know how to use them. Personal fall-arrest systems are required for workers who use boatswain's chairs.

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9. Why do most falls occur when using a descent-control device with a boatswain's chair?

a. Failure of the planking on the boatswain's chair
b. Failure of the descent device locking mechanism
c. Failure of the primary support rope or anchor
d. Failure of the descent device connection

Crane- and Derrick-Suspended Personnel Platforms

For some jobs, adjustable-suspension scaffolds may not be feasible or safe. When there is no other safe way to reach the work area, a crane or a derrick can provide suspended access by hoisting a personnel platform to reach the work area.

Workers rarely fall from suspended personnel platforms. Rather, most fatal accidents happen when the boom or another part of the crane contacts an energized power line. Causes for other fatal or serious-injury accidents include:

- **Instability**: Unstable ground or support surface causes the crane to tip over.
- **Lack of communication**: The crane operator can't see the suspended platform while it is moving.
- **Rigging failure**: Platform loads are not properly rigged.
- **Boom failure**: The weight of the loaded platform exceeds the boom's load limit.

Safe Practices

- Stay within the platform while it's moving.
- Wear a body belt or harness and use a lanyard; attach the lanyard to the lower load block or overhaul ball or to a structural member of the platform.
- Stay in view of the crane operator or signal person while you're on the platform.
- Before leaving the platform for the work area, secure it to the structure.

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10. Why do most fatal accidents occur when operating a crane-suspended personnel platform?

a. The crane contacts an energized power line
b. The personnel platform rigging fails
c. The crane's boom fails
d. The personnel platform is unstable due to wind

Scenario

On August 17, 1998, a 15-year-old window washer's helper (the victim) died after falling 40 feet from the roof of a medical office building. The helper was stationed on the roof to move a window washing carriage and assist the window washer, who worked from a boatswain's chair as he cleaned the windows of the 4-story building.

On the afternoon of the incident, the window washer seated himself in the boatswains' chair and positioned himself over the edge of the roof's parapet. He then "bounced" in the boatswain's chair to make sure it was set to go. Because the carriage was not tied down and did not have counterweights attached, the carriage was pulled over the rooftop's parapet. Both workers had their fall arrest harnesses secured to the carriage. The window washer fell straight down while the helper was pulled from the roof by the carriage and struck the ground head-first. The local emergency medical unit was summoned immediately. The window washer's helper died from his injuries at the scene and the window washer suffered multiple severe injuries.

Recommendations:

- Anyone working on, or from a roof with a fall exposure should be tied off with a safety line. The safety line should be attached to a specifically engineered independent anchorage point.
- All persons who work at heights, should be trained, educated, and knowledgeable in all aspects of the safe use of their tools and equipment and be made aware of all the hazards related to their job.
- Work safety and fall protection plans should be developed and implemented at all work sites.

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- Employers need to effectively supervise and coach employees who have little or no experience in performing high-risk jobs.

11. When working on a roof with a fall exposure, safety lines must be attached to _____.

e. any adapted structural component
f. the primary rope's loop designed for that purpose
g. the personnel platform's carriage
h. an engineered independent anchorage point

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Module 5: Fall Protection Systems

Introduction

If workers will be exposed to fall hazards that you can't eliminate, you'll need to prevent falls from occurring or ensure that if workers do fall, they aren't injured. A fall-protection system is designed to prevent or arrest falls.

Types of Fall Protection Systems

There are seven general fall-protection systems:

- personal fall-arrest system (PFAS) arrests (limits) a fall
- personal fall-restraint system prevents a fall
- positioning-device system positions a worker and limits a fall to 2 feet
- guardrail system prevents a fall
- safety-net system arrests a fall
- warning-line system for roofing work warns a worker of a fall hazard

1. Which one of the following fall protection systems limits a fall to 2 feet?

a. Safety-net system
b. Fall arrest system
c. Positioning-device system
d. Fall restraint system

Identify and Evaluate Fall Hazards

We know that eliminating all fall hazards on the construction site is impossible because they are inherent in the nature of construction work. However, it is possible to mitigate (to become less harmful) many hazards for tasks that must be performed at elevation. To begin the process, identify each worksite fall hazard and analyze each one for the following variables that help to determine the risk that a hazard presents:

- Probability – the likelihood of a fall accident if the employee is exposed to a hazard; and

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- Severity – the magnitude of the injury ranging from no injury to fatal injury;
- You can use a risk matrix to help you determine the degree of risk different worksite task might present. Know that will help you determine appropriate fall-protection systems for your worksite.

Many factors affect the probability of a fall accident and the severity of injury.

- **Scope** – the number of employees performing a task
- **Frequency** – how frequently the task will be performed
- **Rapidity** – how quickly an employee must perform a task
- **Duration** – how long it takes an employee to perform a task
- **Elevation** – the height at which an employee will perform a task
- **Orientation** – the body part(s) that will likely strike a lower surface
- **Equipment** – exposure to sharp points, moving parts, and hazardous energy
- **Materials** – exposure to toxic chemicals, substances, and materials
- **Exertion** – the demand on an employee to exert excessive force (lifting, lowering, pushing, or pulling)
- **Physical environment** – workplace temperature, atmospheres, humidity, illumination, and noise
- **Psychosocial environment** – the psychological stress that causes distraction
- **Response** – the promptness of rescue and quality of treatment of an injury

Keep the following in mind:

- What is the fall distance from the walking/working surface to the next lower level?
- How many workers are exposed to the hazard?
- What tasks and work areas are associated with the hazard?
- How will the workers move - horizontally, vertically, or in both directions - to do their tasks?

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- Are secure anchorages available or can they be easily installed near the hazard?
- Are there other hazards near the work area, such as overhead power lines?
- How will workers be promptly rescued if they are suspended in a personal fall-arrest system?

For more information on determining risk, see OSHAcademy Course 706 Job Hazard Analysis.

2. What are the two primary variables that help determine the risk that a hazard presents?

a. Duration and scope
b. Probability and severity
c. Elevation and exertion
d. Frequency and physical environment

Personal Fall Arrest Systems (PFAS)

A personal fall-arrest system consists of an anchorage, connectors, and a full-body harness that work together to stop a fall and to minimize the arrest force. Other parts of the system may include a lanyard, a deceleration device, and a lifeline.

- Ensure that personal fall arrest systems will, when stopping a fall:
    - Limit maximum arresting force to 1,800 pounds.
    - Be rigged such that an employee can neither free fall more than 6 feet nor contact any lower level.
    - Bring an employee to a complete stop and limit maximum deceleration distance to 3½ feet.
    - Have sufficient strength to withstand twice the potential impact energy of a worker free falling a distance of 6 feet, or the free fall distance permitted by the system, whichever is less
- Remove systems and components from service immediately if they have been subjected to fall impact, until inspected by a competent person and deemed undamaged and suitable for use.

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- Promptly rescue employees in the event of a fall or assure that they are able to rescue themselves.
- Inspect systems before each use for wear, damage, and other deterioration, and remove defective components from service.
- Do not attach fall arrest systems to guardrail systems or hoists.
- Rig fall arrest systems to allow movement of the worker only as far as the edge of the walking/working surface, when used at hoist areas.

3. What is the required height of scaffold top rails, or equivalent guardrail system members above the walking-working level?

a. Between 27 and 33 inches
b. 32 plus or minus 4 inches
c. Between 36 and 42 inches
d. 42 inches plus or minus 3 inches

Body Harness

Body harnesses are designed to minimize stress forces on an employee's body in the event of a fall, while providing sufficient freedom of movement to allow work to be performed. Harnesses, and components must be used only for employee protection (as part of a personal fall arrest system) and not to hoist materials.

Keep the following in mind:

- The harness must be made from synthetic fibers.
- The harness must fit the user. It should be comfortable and easy to adjust.
- According to ANSI/ASSE Z359.1, Safety Requirements for Personal Fall Arrest Systems, Subsystems, and Components, the harness must have an attachment point, usually a D-ring, in the center of the back at about shoulder level. A D-ring may also be used in the front of the harness. However, connection at the front D-ring is limited to systems that restrict free fall distance to 2 ft. or less and limit the maximum fall arrest loads on the front D-ring to 900 lb. of force or less. The D-ring should be large enough to easily accept a lanyard snap hook.

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- Chest straps should be easy to adjust and strong enough to withstand a fall without breaking.
- Use only industrial full-body harnesses (not recreational climbing harnesses).
- The harness must be safe and reliable. It should meet ANSI and CSA standards and the manufacturer should have ISO 9001 certification, which shows the manufacturer meets international standards for product design, development, production, installation, and service.

Body Belts

A body belt is a strap with means both for securing about the waist and for attaching to other components such as a lanyard used with a positioning system while working on a vertical surface, or a travel restraint system when on a horizontal work surface.

Body belts are not acceptable as part of a personal fall arrest system because they impose a danger of internal injuries when stopping a fall. Body belts may only be used as part of a vertical positioning system or a horizontal restraint system.

4. What must a body harness be made of?

a. Organic or synthetic fibers
b. Natural fibers
c. Synthetic fibers
d. Synthetic or natural fibers

The Anchorage

An anchorage is a secure point of attachment for lifelines, lanyards, or deceleration devices. How can you be sure that an anchorage is secure? An anchorage for a personal fall-arrest system must support at least 5,000 pounds. Anchorages that can't support 5,000 pounds must be designed and installed under the supervision of a qualified person and must be able to maintain a safety factor of at least two - twice the impact force of a worker free-falling 6 feet. If you don't know how much weight an anchorage will support, have a qualified person check it before you trust your life to it.

Anchorage strength is critical but is not the only factor to consider. Also, important:

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- Anchorage connector: Unless an existing anchorage has been designed to accept a lanyard or lifeline, you'll need to attach an anchorage connector - a device that provides a secure attachment point. Examples include tie-off adapters, hook anchors, beam connectors, and beam trolleys. Be sure that the connector is compatible with the lanyard or lifeline and appropriate for the work task.

- Attachment point: The anchorage can be used only as the attachment point for a personal fall-arrest system; it can't be used to support or suspend platforms.

- Location: The anchorage should be located directly above the worker, if possible, to reduce the chance of a swing fall.

- Fall distance: Because a personal fall-arrest system doesn't prevent a fall, the anchorage must be high enough above a worker to ensure that the arrest system, and not the next lower level, stops the fall. Consider free-fall distance, lanyard length, shock-absorber elongation, and body-harness stretch in determining the height of an anchorage. Free-fall distance is the distance a worker falls before a personal fall-arrest system begins to stop the fall.

- Connectors: An anchorage, a lanyard, and a body harness are not useful until they're linked together. Connectors do the linking; they make the anchorage, the lanyard, and the harness a complete system. Connectors include carabiners, snap hooks, and D-rings.

- Carabiner: This high-tensile alloy steel connector has a locking gate and is used mostly in specialized work such as window cleaning and high-angle rescue. Carabiners must have a minimum tensile strength of 5,000 pounds.

- Snap hook: A hook-shaped member with a keeper that opens to receive a connecting component and automatically closes when released. Snap hooks are typically spliced or sewn into lanyards and self-retracting lifelines. Snap hooks must be high-tensile alloy steel and have a minimum tensile strength of 5,000 pounds. Use only locking snap hooks with personal fall-arrest systems; locking snap hooks have self-locking keepers that won't open until they're unlocked.

- D-ring: D-rings are the attachment points sewn into a full-body harness. D-rings must have a minimum tensile strength of 5,000 pounds.

- The full-body harness: The full-body harness has straps that distribute the impact of a fall over the thighs, waist, chest, shoulders, and pelvis. Full-body harnesses come in

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different styles, many of which are light and comfortable. Before you purchase any harness, make sure they fit those who will use them, they're comfortable, and easy to adjust. A full-body harness should include a back D-ring for attaching lifelines or lanyards and a back pad for support.

5. How many pounds must an anchorage for a personal fall-arrest system support?

a. At least 2,000 pounds
b. More than 3,000 pounds
c. Up to 4,000 pounds
d. At least 5,000 pound

Lanyards

A lanyard is a specially designed flexible line that has a snap hook at each end. One snap hook connects to the body harness and the other connects to an anchorage or a lifeline. Lanyards must have a minimum breaking strength of 5,000 pounds. They come in a variety of designs, including self-retracting types that make moving easier and shock-absorbing types that reduce fall-arrest forces. Don't combine lanyards to increase length or knot them to make them shorter.

Deceleration Devices

Deceleration devices protect workers from the impact of a fall and include shock-absorbing lanyards, self-retracting lifelines or lanyards, and rope grabs.

Shock-absorbing Lanyard

A shock absorber reduces the impact on a worker during fall arrest by extending up to 3.5 feet to absorb the arrest force. OSHA rules limit the arrest force to 1,800 pounds but a shock-absorbing lanyard can reduce the force even more - to about 900 pounds.

Because a shock-absorbing lanyard extends up to 3.5 feet, it's critical that the lanyard stops the worker before the next lower level. Allow about 20 vertical feet between the worker's anchorage point and the level below the working surface. Always estimate the total distance of a possible fall before using a shock-absorbing lanyard. Example: Lanyard length (6 feet) + deceleration distance (3.5 feet) + worker's height (6 feet) + safety margin (3 feet) = 18.5 vertical feet from anchorage to lower level.

Never use a shock-absorbing lanyard if the shock absorber is even partially extended or if the lanyard has arrested a fall.

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6. How many feet does a shock-absorbing lanyard extend a fall?

a. An additional 4 feet
b. Up to 3.5 feet
c. About 2 feet
d. Less than 5 feet

Self-Retracting Lanyard/Lifeline

Self-retracting lanyards and lifelines offer more freedom to move than shock-absorbing lanyards. Each has a drum-wound line that unwinds and retracts as the worker moves. If the worker falls, the drum immediately locks, which reduces free-fall distance to about 2 feet - if the anchorage point is directly above the worker. Some self-retracting lanyards will reduce free-fall distance to less than one foot. Self-retracting lanyards are available in lengths up to 20 feet. Self-retracting lifelines, which offer more freedom, are available in lengths up to 250 feet.

- Self-retracting lanyards and lifelines that limit free-fall distance to 2 feet or less must be able to hold at least 3,000 pounds with the lanyard (or lifeline) fully extended.

- Self-retracting lanyards that don't limit free-fall distance to 2 feet must be able to hold at least 5,000 pounds with the lanyard (or lifeline) fully extended.

If you use a self-retracting lanyard or lifeline, work below the anchorage to avoid a swing fall. The farther you move away from the anchorage, the farther you will fall and the greater your risk of swinging back into a hard object. Swing falls are hazardous because you can hit an object or a lower level during the pendulum motion.

7. A self-retracting lanyard or lifeline reduces the free-fall distance to about 2 feet if _____.

a. the worker has not extended the lanyard
b. the anchorage point is directly above the worker
c. the work is on a low-slope roof
d. the anchorage attached at shoulder level

Rope Grab

A rope grab allows a worker to move up a vertical lifeline but automatically engages and locks on the lifeline if the worker falls. When using a rope grab, keep the following in mind.

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- Inspect the rope grab before use for distortion, rust, sharp edges and jammed moving parts.
- Make sure the rope grab is compatible with the lifeline (the right rope size or type).
- Attach the rope grab correctly to the lifeline (arrow points up).
- After attaching the rope grab to the lifeline, give it a firm tug to verify it engages.
- Keep the distance on the lanyard between the rope grab and the body harness as short as possible.
- Keep the rope grab as high as possible above the d-ring on the lifeline to shorten the free-fall distance.
- Do not grab the device during a fall.
- Only one person at a time may use the same vertical lifeline.
- If the rope grab arrests a fall, inspect and recertify it for use.

8. After attaching the rope grab to the lifeline, what should you do to test it?

a. Give the rope grab a firm downward tug
b. Grab the device to test its ability to hold
c. Ensure the arrow is pointing down on the device
d. Keep the rope grab at or below the harness d-ring

Lifelines

A lifeline is a cable or rope that connects to a body harness, lanyard, or deceleration device, and at least one anchorage. There are two types of lifelines, vertical and horizontal.

Vertical lifeline: A vertical lifeline is attached to an overhead anchorage and must be connected directly to a worker's full-body harness, lanyard, retractable device, or rope grab; it must have a minimum breaking strength of 5,000 pounds. When a worker needs to move horizontally, however, a vertical lifeline can be hazardous due to the potential for a swing fall - the pendulum motion that results when the worker swings back under the anchor point. A swing fall increases a worker's risk of striking an object or a lower level during the pendulum motion.

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Horizontal lifeline: Unlike a vertical lifeline, the horizontal lifeline stretches between two anchorages. When you connect a lanyard or rope grab to the horizontal lifeline, you can move about freely, thus reducing the risk of a swing fall. However, horizontal lifelines are subject to much greater loads than vertical lifelines. If they are installed incorrectly, horizontal lifelines can fail at the anchorage points. For this reason, horizontal lifelines must be designed, installed, and used under the supervision of a qualified person.

Horizontal lifelines and sag angles: Any load on a horizontal lifeline will cause it to deflect, or sag. The sag angle is a horizontal lifeline's angle of deflection when it's subjected to a load, such as a falling worker. Reducing the sag angle (making a horizontal lifeline too tight) actually increases the force on the line during a fall. As you tighten a horizontal lifeline, you increase the impact load dramatically!

Example: When the sag angle is 15 degrees, the force on the lifeline and anchorages subjected to a load is about 2:1. However, if you decrease the sag angle to 5 degrees, the force increases to about 6:1. To reduce loads on a horizontal lifeline, increase the sag angle or connect to the lifeline with a shock-absorbing lanyard.

9. What advantage does a horizontal lifeline have over a vertical lifeline?

a. Arrest forces are increased during a fall
b. You can easily move horizontally as well as vertically
c. There is a reduced risk of a swing fall
d. A qualified person must supervise its use

Some Do’s and Don’ts

Do's

- Read and follow manufacturer instructions for assembly, use, and care of fall arrest system components.
- After a fall, immediately remove the system from service. Also, have a competent person inspect the anchorage points of the system involved in the fall.
- Inspect fall arrest equipment (as well as all PPE) before and after use.
- When inspecting, consider possible environmental hazards (e.g., heat, chemicals, corrosive environments, high voltage power lines, moving machinery, sharp edges, etc.).

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- Check the sag angle of a horizontal lifeline to estimate arrest forces on the anchorages.
- Think about the potential for a swing fall whenever you connect a lifeline to a personal fall-arrest system.
- Check your fall distance by considering lanyard length (before the shock absorber extends), deceleration distance (shock-absorber extension), worker height, and a safety margin (allow 3 feet).

Don'ts

- Don't use a fall arrest system until you have been properly trained in its use, care, and limitations by a competent person.
- Don't use a fall arrest system unless you have developed a rescue plan. The longer you hang in the air, the more serious will be our injuries.
- Don't tie knots in rope lanyards and lifelines; knots can reduce strength by 50%.
- Don't tie lifelines or lanyards directly to I-beams; the cutting action of beam edges can reduce the rope's strength by 70%.
- Never attempt to use makeshift components that have not been specifically designed for fall arrest applications.

10. Why shouldn't you tie knots in rope lanyards or lifelines?

a. It can weaken the lanyard by 70%
b. It can slip through the d-ring during use
c. It can get caught and cause a fall
d. It reduces the strength of the lanyard by 50%

Scenario

Five roofing-company workers had been removing cedar shingles and replacing them with plywood sheeting and composition roofing at a two-story home on an afternoon in mid-January.

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The crew had stopped work for lunch and returned to work about 1:30 p.m. While four of the crew went up on the roof, the victim remained on the ground to push plywood sheets up an extension ladder to crew members on the roof. When all the plywood sheets were on the roof, the victim climbed the ladder and got on the roof. Then he bent down near the top of the ladder, apparently to adjust it.

Another worker on the roof heard a loud noise, rushed over to the ladder, and discovered that the victim had fallen 17 feet to the ground.

The workers climbed down to assist the victim and the supervisor called 911 on his cell phone. The workers administered first aid and immobilized the victim's neck. EMTs took the victim to a hospital where he died later that day of traumatic head injuries.

Findings: The victim, who was hired the day of the accident, had no fall-protection training or instruction in ladder use. Workers at the site had fall-protection equipment but were not using it according to the manufacturer's instructions; the victim was not using the equipment. The roof edge was more than 17 feet above the ground and the ladder was not tied off.

11. What is the most common root cause of roofing construction accidents?

a. Lack of enforcement of safety rules
b. Inadequate safety training
c. Poor worker supervision
d. Lack of employee involvement in safety

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Module 6: Fall Protection Systems (Continued)

Horizontal Travel-Restraint Systems

A travel restraint system defined in 1910.140(b) means a combination of an anchorage, anchorage connector, lanyard (or other means of connection), and body support that an employer uses to eliminate the possibility of an employee going over the edge of a walking-working surface. The system does not retrain a fall: it eliminates the possibility.

A travel restraint (tether) line defined in 1910.140(b) means a rope or wire rope used to transfer forces from a body support to an anchorage or anchorage connector in a travel restraint system. Travel restraint lines must be capable of sustaining a tensile load of at least 5,000 pounds (22.2 kN).

Anchorages, except window cleaner belt anchors must be capable of supporting at least 5,000 pounds (22.2 kN) for each employee attached; or designed, installed, and used, under the supervision of qualified person, as part of a complete personal fall protection system that maintains a safety factor of at least two.

1. Which fall protection system eliminates the possibility of falling over the edge of a walking-working surface?

a. Travel restraint system
b. Fall arrest system
c. Positioning system
d. Guardrail system

Vertical Positioning-Device Systems

A positioning device is defined in 1926.500(b) means a body belt or body harness system rigged to allow an employee to be supported on an elevated vertical surface, such as a wall, and work with both hands free while leaning. The system is used only when the work surface it vertical.

These devices are designed specifically to stop a worker from falling from a static, head-up position. Under 1926.502(e), a positioning device must limit the fall to 2 feet.

The components of a positioning-device system - anchorage, connectors, and body support.

- **Anchorage:** The positioning devices must be secured to an anchorage capable of supporting at least twice the potential impact load of an employee's fall or 3,000 pounds (13.3 kN), whichever is greater.

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- **Connectors:** Connectors must have a minimum strength of 5,000 pounds. Snap hooks and D-rings must be proof-tested to a minimum load of 3,600 pounds without deforming or breaking.
- **Body support:** A body belt is acceptable as part of a positioning-device system. However, it must limit the arresting force on a worker to 900 pounds and it can only be used for body support.

A full-body harness is also acceptable and must limit the arrest force to 1,800 pounds. Belts or harnesses must have side D-rings or a single front D-ring for positioning.

Questions:

1. Can an employee on an aerial lift use a body belt if it is used to restrain the employee from falling rather than arresting a fall, such as with a personal fall arrest system?
2. Can an employee on an aerial lift use a body belt if it is used as part of a positioning device?

Answers:

The difference between a restraint system and a positioning device:

A restraint system is one that prevents a worker from falling any distance. In contrast, for construction work covered by 29 CFR Part 1926 Subpart M (Fall Protection), a "positioning device" is defined in 1926.500(b) as "a body belt or body harness system rigged to allow an employee to be supported on an elevated **vertical** surface, such as a wall, and work with both hands free while leaning." These devices are designed specifically to stop a worker from falling from a static, head-up position. Under 1926.502(e), a positioning device must limit the fall to 2 feet.

Appropriate body belt use in aerial lifts

An employee can use a body belt when the body belt is used in a restraint system -- a system that prevents the worker from being exposed to any fall.

A body belt is permitted to be used in a positioning device. However, since positioning devices are defined as systems used to protect workers on vertical surfaces, positioning devices are not permitted in bucket-type aerial lifts, where the work surface is horizontal working from an aerial lift. An employee must be protected by a personal fall arrest system. Under 1926.502(d), body belts are not permitted to be used in a personal fall arrest system, and the requirement is not negated by the use of a positioning device that limits any fall to 2 feet.

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Sources: OSHA Interpretation Letter May 11, 2001. 1926.453(a) and (b)(2), 1926.500(b), 1926.502(d) and (e); aerial lifts; fall arrest systems; positioning devices

2. Which fall protection system must limit a fall to no more than two feet?

a. Fall restraint system
b. Fall arrest system
c. Positioning system
d. Guardrail system

Guardrail Systems

A guardrail system consists of a top rail, midrail, and intermediate vertical member. Guardrail systems can also be combined with toeboards that prevent materials from rolling off the walking/working surface.

If a guardrail system is required, be sure to comply with the following provisions:

- Top edge height of top rails, or equivalent guardrail system members, must be 42 inches plus or minus 3 inches above the walking/working level, except when conditions warrant otherwise and all other criteria are met (e.g., when employees are using stilts, the top edge height of the top rail must be increased by an amount equal the height of the stilts).

- Midrails, screens, mesh, intermediate vertical members, or equivalent intermediate structures, must be installed between the top edge and the walking/working surface when there is no wall or other structure at least 21 inches high.

    - Midrails must be midway between the top edge of the guardrail system and the walking/working level.

    - Screens and mesh must extend from the top rail to the walking/working level, and along the entire opening between rail supports.

    - Intermediate members (such as balasters) between posts must be no more than 19 inches apart.

    - Other structural members (such as additional midrails or architectural panels) must be installed so as to leave no openings wider than 19 inches.

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- Guardrail systems must be capable of withstanding at least 200 pounds of force applied within 2 inches of the top edge, in any direction and at any point along the edge, and without causing the top edge of the guardrail to deflect downward to a height less than 39 inches above the walking/working level.

- Midrails, screens, mesh, and other intermediate members must be capable of withstanding at least 150 pounds of force applied in any direction at any point along the midrail or other member.

- Guardrail systems must not have rough or jagged surfaces that would cause punctures, lacerations, or snagged clothing.

- Top rails and midrails must not cause a projection hazard by overhanging the terminal posts.

3. What is the required height of scaffold top rails, or equivalent guardrail system members above the walking-working level?

a. Between 27 and 33 inches
b. Between 30 and 36 inches
c. Between 36 and 42 inches
d. Between 39 and 45 inches

Safety Net Systems

Many times, the nature and location of the work will dictate the form that fall protection takes. If the employer chooses to use a safety net system, he must comply with the following provisions:

- Safety nets must be installed as close as practicable under the surface on which employees are working, but in no case more than 30 feet below.

- When nets are used on bridges, the potential fall area must be unobstructed.

- Safety nets must extend outward from the outermost projection of the work surface as follows:

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Vertical Distance From working level to horizontal plane of net	Horizontal Distance From outer edge of net to the edge of the working surface
Up to 5 feet	8 feet
5 to 10 feet	10 feet
More than 10 feet	13 feet

- Safety nets must be installed with sufficient clearance to prevent contact with the surface or structures under them when subjected to an impact force equal to the drop test described below.
- Safety nets and their installations must be capable of absorbing an impact force equal to the drop test described below.
- Safety nets and safety net installations must be drop-tested at the jobsite:
    - after initial installation and before being used
    - whenever relocated
    - after major repair
    - at 6-month intervals if left in one place
- The drop test consists of a 400 pound bag of sand 28-32 inches in diameter dropped into the net from the highest surface at which employees are exposed to fall hazards, but not from less than 42 inches above that level.
- When the employer can demonstrate that it is unreasonable to perform the drop-test described above, the employer or a designated competent person will certify that the net and net installation have sufficient clearance and impact absorption by preparing a

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certification record prior to the net being used as a fall protection system. The certification must include:

  - identification of the net and net installation
  - date it was determined the net and net installation were in compliance
  - signature of the person making the determination and certification

- The most recent certification record for each net and net installation must be available at the jobsite for inspection.
- Safety nets must be inspected for wear, damage, and other deterioration at least once a week, and after any occurrence which could affect the integrity of the system.
- Defective nets shall not be used, and defective components must be removed from service.
- Objects which have fallen into the safety net, such as scrap pieces, equipment, and tools, must be removed as soon as possible from the net and at least before the next work shift.
- Maximum mesh size must not exceed 6 inches by 6 inches. All mesh crossings must be secured to prevent enlargement of the mesh opening, which must be no longer than 6 inches, measured center-to-center.
- Each safety net, or section thereof, must have a border rope for webbing with a minimum breaking strength of 5,000 pounds.
- Connections between safety net panels must be as strong as integral net components and must not be spaced more than 6 inches apart.

4. What is the maximum distance safety nets may be installed under the work surface?

a. 10 feet
b. 15 feet
c. 30 feet
d. 35 feet

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Warning Line Systems for Roofing Work

Roofing work refers to hoisting, storing, applying, and removing roofing materials and equipment; it includes work on related insulation, sheet metal, and vapor barriers, but does not include the construction of the roof deck or leading-edge work. A warning-line system for roofing work consists of ropes, wires or chains, and supporting stanchions that mark off an area within which roofing work can be done without guardrails, personal fall-arrest systems, restraint systems, or safety nets. Warning-line systems can only be used for roofing work on roofs that have slopes of 2:12 or less, vertical to horizontal. The purpose of the line is to warn roofers that they are near an unprotected edge.

The warning line must be at least 6 feet from an unprotected edge and meet the following criteria:

- be flagged at least every 6 feet with high-visibility material
- be rigged so the line is 34 to 39 inches from the walking/working surface
- have a minimum tensile strength of 500 pounds and don't use plastic caution tape for a warning line
- Be attached to each stanchion so that tension on one section of the line will not cause an adjacent stanchion to tip over. Stanchions must be able to support a force of at least 16 pounds applied horizontally in the direction of the roof edge without tipping over.

Those who do roofing work between the warning line and an unprotected roof edge must be protected with personal fall-arrest systems, restraint systems, guardrail systems, safety monitoring systems, or safety nets.

5. How far must a warning line be placed from an unprotected edge?

a. At least 6 feet
b. Up to 8 feet
c. Ten feet or more
d. Over 15 feet

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Slide-Guard Systems

A slide-guard system prevents workers from sliding down a sloped roof. The system consists of a slide guard (typically 2-by-6-inch lumber) and at least two roof brackets and must be installed under the supervision of a competent person. Roof brackets are available from roofing-equipment suppliers. A slide-guard system can also be made at the work site without manufactured roof brackets. Slide-guard systems cannot be the only means of fall protection on roofs with a ground-to-eave height greater than 25 feet.

When are Slide Guards Allowed?

Slide guards are not permitted in lieu of conventional fall protection methods during roofing work (removal, repair, or installation of weatherproofing roofing materials, such as shingles, tile, and tar paper). However, slide guards may be used as part of a written, site-specific fall protection plan that meets the requirements of 29 CFR 1926.502(k) only if the employer can demonstrate that the use of conventional fall protection would be infeasible or would create a greater hazard.

6. In addition to having a written site-specific fall protection plan, what requirement must an employer meet before being allowed to use a slide-guard system?

a. The employer must demonstrate those slide guards would allow for greater movement of workers horizontally
b. The employer must show that using a fall arrest system would be difficult
c. The employer must verify work is being performed on high-pitch a composite roof
d. The employer must prove the use of conventional fall protection is infeasible or more hazardous

Safety Monitoring

A safety-monitoring system involves a competent person who is responsible for recognizing and warning employees of fall hazards. The method relies on a person, rather than a mechanical system, to warn roofers if they are in danger of falling.

Each employee engaged in roofing activities on low-slope roofs, with unprotected sides and edges 6 feet (1.8 m) or more above lower levels must be protected from falling by:

- guardrail systems, safety net systems, personal fall arrest systems, or
- a combination of warning line system and:

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- a guardrail system,
- a safety net system,
- a personal fall arrest system, or
- a safety monitoring system.

On roofs 50-feet (15.25 m) or less in width (see 1926 Subpart M Appendix A), the use of a safety monitoring system alone [i.e. without the warning line system] is permitted.

The safety monitor's responsibilities include:

- recognizing fall hazards;
- warning employees when they are unaware of hazards or aren't working safely;
- staying on the same walking/working surface as the workers to see them and to communicate with them while they are working; and
- avoiding any other work or distracting activity while monitoring the workers.

Only those who are doing roofing work are permitted in the area controlled by the safety monitor. Mechanical equipment can't be used or stored in the area.

7. Which one of the following actions violates a responsibility assigned to safety monitors?

a. Avoiding other work
b. Monitoring employees from a lower walking-working surface
c. Ensuring a warning line system is installed as required
d. Warning employees of recognized hazards

Catch Platforms

Catch platforms, which consist of a stable platform and an attached standard guardrail, can protect roofers when other systems or methods are not feasible. Platform guidelines:

- The platform should not be more than 18 inches below the eave line of the roof.
- The platform should extend horizontally at least 2 feet beyond the eave line of the roof.

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- The platform must have a standard guardrail and toeboard. The top guardrail should rise substantially (at least 12 inches) above the eave line of the roof. Install intermediate rails or a solid barrier between the top rail and the platform to prevent a worker from sliding under the top rail.

- The platform must be able to support employees and materials that fall on to it.

8. How far should the platform on a catch platform extend horizontally beyond the eave line of the roof?

a. At least 2 feet
b. More than 3 feet
c. Up to 4 feet
d. Between 3 and 4 feet

Hole Covers

Simple and effective when they're properly installed, rigid covers prevent workers from falling through skylights or temporary openings and holes in walking/working surfaces.

Safety criteria for covers:

- Covers will support at least twice (2 times) the maximum expected weight of workers, equipment, and materials. Skylights are not considered covers unless they meet this strength requirement.

- They are secured to prevent accidental displacement.

- They have full edge bearing on all four sides.

- They are painted with a distinctive color or marked with the word HOLE or COVER.

Fences and Barricades

Fences and barricades are warning barriers. They are usually made from posts and wire or boards that keep people away from hazards such as wells, pits, and shafts.

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9. How much weight must a hole cover support?

a. The anticipated weight of workers, equipment, and materials
b. The potential maximum expected weight of workers, equipment, and materials
c. At least twice the maximum expected weight of workers, equipment, and materials
d. At least 3,000 pounds or the expected weight of workers, equipment, and materials

Protecting Workers from Falling Objects

You need to protect yourself from falling when you work on an elevated surface and be aware of those working above or below you. Protect yourself and others from falling objects with one of the following methods:

- Canopies: Make sure canopies won't collapse or tear from an object's impact.
- Toeboards: Toeboards must be least 3½ inches high and strong enough to withstand a force of at least 50 pounds applied downward or outward.
- Panels and screens: If you need to pile material higher than the top edge of a toeboard, install panels or screens to keep the material from dropping over the edge.
- Barricades and fences: Use them to keep people away from areas where falling objects could hit them.

When doing overhand bricklaying, keep materials and equipment (except masonry and mortar) at least 4 feet from the working edge. When doing roofing work, keep materials and equipment at least 6 feet from the roof edge unless there are guardrails along the edge. All piled, grouped, or stacked material near the roof edge must be stable and self-supporting.

10. When working on a roof, how far should materials and equipment be kept from the roof edge?

a. Between 3 and 4 feet depending on the pitch of the roof
b. More than 4 feet when working on a pitched roof
c. At least 6 feet when the edge has no guardrail with toeboards
d. At least 2 feet if the material is stable

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Module 7: Fall Protection Training

The Importance of Training

Workers need to know about construction worksite hazards to which they may be exposed, how to recognize the hazards, and how to minimize their exposure. The best way for them to learn is through training. Training ensures that they know about the hazards and can demonstrate how to protect themselves from falling.

The KSA Education Process

How can the employer give employees the most effective training to perform safely? We call it the KSA Education Process, and it has basic integrated educational strategies:

1. **Instruction transfers Knowledge:** This is where the educational process begins. Employees must know something before they can do something. Workers are instructed on safety policies, procedures, practices, and rules, and why they are important.

2. **Training develops Skills:** Once we know something, we can focus on learning how to do something. The best way to do that is through hands-on practice in the learning environment. Giving trainees an opportunity to practice the use of equipment, procedures, and practices is required for most fall protection training.

3. **Experience improves Abilities:** Employees gain experience outside the classroom on the job, where the "real education" occurs. Only with experience will employees improve our overall performance.

Some employers assume that they can train their employees simply by showing them a fall-protection training video or online course (like this one!). But videos, lectures, or online courses alone are not adequate because they do not provide the "hands-on" component of the training. That's why most safety training must be conducted by an employer-certified competent person in-house.

1. Which component of most fall protection training is required to ensure employees have adequate knowledge, skills, and abilities (KSAs)?

a. Common sense
b. Instruction
c. Experience
d. Hands-on practice

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Employers: Your Responsibility

If you're an employer, you're responsible for ensuring your employees can recognize fall hazards and that they know how to protect themselves before they're exposed to the hazards. You can't assume they know how to protect themselves from falls. If they're starting work on a new site, for example, they might not recognize fall hazards or know how to protect themselves unless you train them.

Required Training

Workers who could be exposed to fall hazards must be trained to recognize the hazards and to know the procedures that minimize the hazards.

OSHA requires the following minimum fall-protection training:

Fall Protection -- Training Requirements in Construction 1926.503(a)(1) and (2)(ii) through (vii)

(a) Training Program:

(1) The employer shall provide a training program for each employee who might be exposed to fall hazards. The program shall enable each employee to recognize the hazards of falling and shall train each employee in the procedures to be followed in order to minimize these hazards.

(2) The employer shall ensure that each employee has been trained, as necessary, by a competent person qualified in the following areas:

(i) The nature of fall hazards in the work area;

(ii) The correct procedures for erecting, maintaining, disassembling, and inspecting the fall protection systems to be used;

(iii) The use and operation of guardrail systems, personal fall arrest systems, safety net systems, warning line systems, safety monitoring systems, controlled access zones, and other protection to be used;

(iv) The role of each employee in the safety monitoring system when this system is used;

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(v) The limitations on the use of mechanical equipment during the performance of roofing work on low-slope roofs;

(vi) The correct procedures for the handling and storage of equipment and materials and the erection of overhead protection; and

(vii) The standards contained in this subpart.

Additional Training Guidelines

ANSI Z359.2, Minimum Requirements for a Comprehensive Managed Fall Protection Program, sets goals for best practices in safety training throughout the industry. It incorporates by reference ANSI/ASSP Z490.1, Criteria for Accepted Practices in Safety, Health and Environmental Training.

2. Who is held responsible and accountable by OSHA for making sure employees can recognize fall hazards and how to protect themselves from those hazards?

a. Each employee
b. Safety committees
c. The competent person on site
d. The employer

Who Can Train

Safety trainers should be competent in developing and presenting safety training courses. Trainers can gain competency by achieving an appropriate level of technical knowledge, skills, and abilities (KSAs) in the fall protection subjects they teach. They can gain these skills through training, continuing education and, of course, on the job experience.

Instructors should be deemed competent on the basis of the criteria below:

1. **They have previous documented experience** in their area of instruction. Job descriptions, performance appraisals, and statements by co-workers or managers would help document previous experience.

2. **Successful completion of a construction "train-the-trainer" program** specific to the topics they will teach. A typical train-the-trainer course will discuss best practices in adult training principles, and provide an opportunity to practice presenting instruction

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and training in the subject area they will be expected to train. See OSHAcademy Train the Trainer Program.

3. **They possess an evaluation of instructional competence** by a training manager or other competent person.

It's important for the employer to document trainer competency by maintaining course completion certificates, experience records, licensing, and other documents. The methods used to document trainer competency is left to the discretion of the employer.

The trainer must know and be able to explain the following topics, including:

- the nature of fall hazards at the work site;
- procedures for erecting, maintaining, and disassembling fall protection systems;
- how to use and operate fall-protection systems;
- the role of each employee who may be affected by a safety-monitoring system;
- restrictions that apply to mechanical equipment used during roofing work;
- procedures for handling and storing materials and for erecting protection from falling objects;
- requirements detailed in OSHA standards; and
- company fall protection policies, procedures, practices, and rules.

3. Which of the following best helps to determine safety trainer competency?

a. Position within the company
b. Knowledge, skills, and abilities (KSAs)
c. Years employed by the company as a supervisor
d. Membership on the company's safety committee

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When to Train

Initial training: Employees must be trained before they begin tasks that could expose them to fall hazards or before they use fall-protection systems.

Retraining: Employees must be retrained when you have reason to believe they don't recognize fall hazards, when they don't follow safe practices for using fall-protection systems, and when changes at the construction worksite or in the fall-protection systems used make their previous training obsolete.

Certification

OSHA expects safety training that includes potentially hazardous procedures and practices to include hands-on practice, demonstration, and evaluation to ensure the employee has gained adequate knowledge, skills, and abilities (KSAs) and is qualified to use the fall protection equipment and complete tasks in the work environment.

Before the employer can certify an employee as "fully qualified," the employee should be evaluated as follows:

1. Initial qualification: To be initially qualified, the employee must be evaluated by the trainer for adequate KSAs in the learning environment; and

2. Full qualification: To be fully qualified, the employee must be evaluated by a competent person for adequate KSAs in the work environment.

The employer should develop a written certification record to document any training that requires employees to know and use procedures and practices for dangerous tasks. See sample certification record.

4. Since fall protection training includes potentially hazardous procedures and practices, OSHA expects it to include _____.

a. hands-on practice, demonstration, and evaluation
b. at least 10 hours of classroom instruction
c. three days of actual experience in the field
d. a test before and after the training has been completed

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Model Training Strategy

The "show and tell" model for on-the-job training (OJT) has been, and is still, the best method for training specific fall-protection safety procedures. Measurement knowledge and skills occurs throughout the OJT process while keeping the employee safe from injury while learning. If, in using this training method, the employee is not exposed to hazards that could cause serious injury, you may be able to delete step 3. Otherwise, do not skip a step.

Step 1: Introduction- The instructor tells the trainee about the training. At this time, the instructor emphasizes the importance of the procedure to the success of the production/service goals, invites questions, and emphasizes accountability.

Step 2: Instructor show and tell- The instructor demonstrates the process. The instructor first explains and demonstrates safe work procedures associated with the task. In this step the trainee becomes familiar with each work practice and why it is important.

Step 3: Instructor show and ask- The trainee tells the instructor how to do the procedure, while the instructor does it. This step is actually optional. It's important to include this step if injury is possible. There is an opportunity for the instructor to discover whether there were any misunderstandings but protects the trainee because the instructor still performs the procedure.

5. What is the best method for training specific fall-protection safety procedures?

a. Classroom training
b. Interactive instruction with discussion
c. The "show and tell" model
d. Instructor-led e-learning

Step 4: Trainee tell and show- Now it's the trainee's turn. To further protect the employee, the Instructor must give permission for the trainee to perform each step. The trainee carries out the procedure but remains protected because the trainer explains the process before actually performing the procedure.

Step 5: Conclusion- The instructor recognizes accomplishment, reemphasizes the importance of the procedure, and how it fits into the overall process. The instructor also reviews the natural consequences (the injury/illness) and system consequences (reward/discipline) related to performance.

Step 6: Document- The trainee certifies (1) training accomplished, (2) questions were answered, (3) opportunities provided to do procedure, (4) accountabilities understood, and (5)

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intent to comply. The instructor certifies that the trainee has (6) demonstrated adequate knowledge and skill to complete the procedure.

6. In step 4 of the Show and Tell method, what should trainees do before they demonstrate a step?

a. Pass the written test prior to performing the task
b. Get permission to do the step
c. Promise to comply with safety rules
d. Ask the trainer if the step is next

What's wrong with this picture?

Take a look at the photo below. See if you can determine the safety hazards associated with each number in the photo. Go online and click on the photo to get the answers.

Photo: OSHA Training Institute, Southwest Education Center

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Module 8: Inspection and Maintenance

Caring for Equipment

When you use ladders, scaffolds, aerial lifts, and fall-protection systems you expect to get your job done safely. But do you pay attention to the condition of the equipment? Inspect the equipment frequently, keep it clean, store it properly, and it won't let you down.

Inspecting Systems

It is very important that you inspect the components of personal fall-arrest, restraint, or positioning-device systems for damage or excessive wear before and after each use. Replace any component that looks damaged. Don't use a personal fall-arrest system that has arrested a fall unless a competent person has determined that the system is safe to use.

1. How often should you inspect fall protection equipment?

a. At least weekly or more often
b. Whenever the equipment appears defective
c. Before and after each use
d. Prior to OSHA inspections

Harness, Lifeline, and Anchorage

Inspect these components regularly.

Review the table below that highlights what to look for:

Component	What to look for
Harness webbing	Frayed edges, broken fibers, pulled stitches, cuts, burns and chemical damage.
Harness D-rings	Cracks, breaks, and rough or sharp edges; the D-ring should pivot easily.
Harness buckles	Excessive wear, frayed or cut fibers, broken stitching.

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Harness grommets	Loose, bent, or broken grommets, and punched holes not made by the manufacturer.
Lifelines	Wear or deterioration.
Anchorages and anchorage connectors	Look for abrasion and damaged threads or swages. Inspect stitching and loops on synthetic slings for cuts, cracks, or frayed and broken stitching. Look for excessive kinks or damaged steel fibers.

2. When should you replace a harness D-ring?

a. If it has cracks
b. If edges are smooth
c. If it pivots easily
d. If it has broken fibers

Snaphooks

Look for cracks, excessive wear, and corrosion. The snaphooks should open easily and close firmly. Keeper locks must prevent the keeper from opening when it's closed.

Lanyards

Type of lanyard	What to look for
Wire rope lanyard	Cuts, frayed strands, or excessive wear.
Web lanyard	Cuts, discoloration, cracks, frayed or broken stitching.
Rope lanyard	Frayed or cut fibers. The entire length of the rope should have the same diameter.

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Shock-absorbing lanyard	Cuts, discoloration, cracks, frayed or broken stitching. Remove a lanyard from service if any part of the warning label is exposed.

3. When should you replace a snaphook?

a. If it opens easily
b. If it has smooth edges
c. If it has corrosion on it
d. If it closes firmly

Self-retracting Lifelines

Look for cuts, frayed strands, or excessive wear in the line and damage to the housing. If the unit needs service, check the manufacturer's recommendations. Don't try to repair it yourself.

See **more information** about PFAS inspection and maintenance from Miller Fall Protection.

See **more information** about PFAS inspection and maintenance from Miller Fall Protection.

Guardrail Systems

Frequently inspect manila, plastic, or synthetic rope used for top rails or midrails to ensure that the rope meets the minimum strength and rail height requirements. [See 1926.502(b)]

Safety-net Systems

Inspect safety nets for damage or deterioration weekly and after any event that could damage them. Remove defective components from service.

4. When should you replace a self-retracting lifeline?

a. The housing is visible
b. It has strands
c. You see normal wear
d. There are cuts in the line

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Ladders

Remember, not just anyone can or should inspect ladders or fall protection equipment. A competent person must inspect ladders periodically. He or she must also inspect them immediately after any event that could damage them.

General Ladders: When inspecting ladders, generally look for loose steps or rungs (considered loose if they can be moved at all with the hand), loose nails, screws, bolts, or other metal parts. Look for cracked, spilt, or broken uprights, braces, or rungs, slivers on uprights, rungs, or steps. Also look for damaged or worn non-slip bases.

Step Ladders: On step ladders, make sure they are not wobbly (from side strain), have loose, bent or broken hinge spreaders, or loose hinges. Make sure the stop on hinge spreaders are not broken. Finally make sure the steps are not broken, split or worn.

Extension Ladders: On extension ladders, make sure the extension locks are not loose, broken, or missing. Make sure locks seat properly while extended, and make sure the rope is not worn, rotted, cut, or defective in any way.

5. When should you take a ladder out of service?

a. When you discover straight and strong uprights
b. When you can move the steps or rungs
c. When you see smooth steps with no slivers
d. When it has functional non-slip bases

Scaffolds

A competent person must inspect a scaffold and its components after it has been erected, before each shift, and after any event - including severe weather - that could damage it. The inspection should include the foundation, platform, guardrails, and access areas.

Suspension Scaffolds

A competent person must inspect suspension ropes before each shift and after any event that could damage them. Inspect and tighten wire rope clips to the manufacturer's recommendations at the start of each shift. Inspect manila or synthetic rope used for toprails or midrails frequently to ensure that it meets the minimum strength and rail height requirements. [See 1926.502(b)].

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6. When must a competent person inspect a scaffold and its components?

e. After it has been erected
f. After each shift
g. Before severe weather strikes
h. Only before OSHA inspects the scaffold

Crane- and Derrick-Suspended Personnel Platforms

- Immediately after a trial lift, a competent person must inspect the rigging, personnel platform, and the base that supports the crane or derrick.
- A competent person must inspect the platform and rigging immediately after they have been proof tested.

Summary: Inspecting, Cleaning, and Storing Fall Protection Equipment

- Always follow manufacturers' instructions and warnings.
- Always inspect equipment before using it. Look for damaged or missing parts. Labels, warnings, and other instructions should be readable.
- If equipment looks like it needs repair, remove it from service and have a competent person examine it.
- Have a competent person inspect equipment regularly.
- Mark equipment with a unique code or item number. Identification numbers make it easier to keep track of the equipment and to document maintenance or repair.
- Wash synthetic rope and body harnesses in soapy water to remove dirt; rinse them with clean water. Air-dry at room temperature. Don't use cleaning solvents; solvents can damage synthetic material.
- Don't lubricate moving parts unless the manufacturer requires it; lubricants attract dirt.
- Don't remove information labels and warnings; make sure they're still legible after cleaning.

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- Follow manufacturer's instruction for storing equipment.
- Store equipment in an area that is clean, dry, and moisture-free; avoid excessive heat, light, oil, and corrosive chemicals.

7. When must a crane or derrick that is lifting a personnel platform be inspected?

a. Before proof testing
b. Before the trial lift
c. After personnel are lifted
d. After the trial lift

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Module 9: Rescue at Height

Prompt Rescue Required

The best strategy for protecting workers from falls is to eliminate the hazards that cause them. When you can't eliminate the hazards, you must protect workers with an appropriate fall-protection system or method. If a worker is suspended in a personal fall-arrest system, you must provide for a prompt rescue using one of the following basic methods:

1. **Self-rescue** - the preferred method using techniques to relieve pressure on legs.

2. **Assisted-rescue** - if self-rescue is not possible, one or more trained rescuers with appropriate equipment perform assisted rescue of the worker.

Prompt rescue of a worker suspended by a fall arrest system is not defined by OSHA in 1926.502. However, OSHA does state that, in this case, the word "prompt" requires that the rescue be performed quickly -- in time to prevent serious injury to the worker. (OSHA Letter of Interpretation 2003-12-18)

A worker suspended in a harness after a fall can lose consciousness within minutes if the harness puts too much pressure on arteries. The Air force studied how long a physically fit person could hang in a full-body harness without extreme discomfort and found that the average times were between 17 and 28 minutes. However, tolerance varies greatly from person to person, and in fact, suspension trauma can occur in as little as 10 minutes.

Check out this great resource, Fall Protection: Responding to Emergencies - WISHA, that covers the various techniques of self-rescue and aided rescue after a worker falls using a personal fall-arrest system (PFAS).

1. If a worker is suspended in a personal fall-arrest system, you must provide for a prompt rescue. "Prompt" means _____.

a. within 15 minutes
b. as soon as possible
c. in time to prevent serious injury
d. when emergency services arrive

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Developing an Emergency-Response Plan

The following guidelines will help you develop a rescue plan for responding promptly to emergencies at height. As we mentioned earlier, responding promptly means to rescue the worker before serious injury while suspended can occur.

- **Effective rescue plans don't have to be elaborate:** Your plan should show that you've thought about how to eliminate and control fall hazards and how to respond promptly if something goes wrong.

- **Get others involved in planning:** When workers who use fall protection equipment, especially competent persons, participate in developing the rescue plan, they'll contribute valuable information, take the plan seriously, and be more likely to respond effectively during an emergency.

- **Develop the Plan:** To develop an effective rescue plan, complete the following sequence of actions:

  1. identify the types of falls that could occur at your site;

  2. establish a communications plan and chain of command;

  3. establish procedures for immediately responding to falls and other accidents;

  4. identify critical resources and rescue equipment; and

  5. train on-site responders; and

  6. regularly practice rescue methods.

2. What is the first step in developing an effective rescue plan for workers who fall while using a fall-arrest system?

a. Establish a communications plan and chain of command
b. Identify critical resources and rescue equipment
c. Identify the types of falls that could occur at your site
d. Train on-site responders

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- Identify emergencies that could affect your construction worksite. Identify any event that could threaten worker safety or health. Two examples:
    - worker suspended in a full-body harness after a fall
    - worker on a scaffold who contacts an overhead power line
- Identify critical resources and rescue equipment. Prompt rescue won't happen without trained responders, appropriate medical supplies, and the right equipment for the emergency.
    - First-aid supplies: Every work site needs medical supplies for common injuries. Does your site have a first-aid kit for injuries that are likely to occur?
    - Store the supplies in clearly marked, protective containers and make them available to all shifts.
- Rescue equipment: Identify on-site equipment that responders can use to rescue a suspended worker.
    - Extension ladders and mobile lifts are useful and available at most sites.
    - Determine where and how each type of equipment would be most effective during a rescue.
    - Think about seasonal and environmental conditions and how they may affect rescue equipment and those who use it. Equipment that works for summer rescues may not work for winter rescues.

The most important question to ask about the rescue equipment is if it will permit rescuers to reach a fall victim. If it can't reach the victim, it's useless. It's also important to make sure it's available when rescuers need it and that rescuers know how to use it. Questions to ask include:

- Will your longest ladder reach a suspended worker?
- If not, what equipment will reach the worker?
- When equipment is needed for a rescue, will workers know where it is and how to use it?

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3. What is the most important question to ask about the rescue equipment when responding to a fall?

a. Can it reach the fall victim?
b. Do the employees know how to use it?
c. Is the equipment available?
d. Is the equipment needed for the rescue?

- Train on-site responders: An effective emergency-response plan ensures that on-site responders know emergency procedures, know how to use available rescue equipment, and - if necessary - know how to contact off-site responders. Workers who use personal fall-arrest systems and who work alone must know how to rescue themselves. Those who work at a remote site may need a higher level of emergency training than those who work near a trauma center or a fire department.

- Establish a chain of command: All workers must know their roles and responsibilities during an emergency. A chain of command links one person with overall responsibility for managing an emergency to those responsible for carrying out specific emergency-response tasks. Make sure that back-up personnel can take over when primary responders aren't available.

- Establish procedures for responding to emergencies. Procedures are instructions for accomplishing specific tasks. Emergency procedures are important because they tell workers exactly what to do to ensure their safety during an emergency. Your emergency-response plan should include the following procedures - preferably in writing - that describe what people must know and do to ensure that a fallen worker receives prompt attention:

    - how to report an emergency

    - how to rescue a suspended worker

    - how to provide first aid

- After an emergency, review the procedures; determine if they should be changed to prevent similar events and revise them accordingly.

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4. What must a lone worker using a fall-arrest system know how to do?

a. Keep themselves mentally stimulated so they don't fall asleep on the job
b. Rescue themselves
c. Contact off-site emergency responders
d. Inspect and maintain their harness

Responding to Falls

When an emergency occurs

- First responders should clear a path to the victim. Others should direct emergency personnel to the scene. You can use 911 for ambulance and medical service; however, most 911 responders are not trained to rescue a worker suspended in a personal fall-arrest system. Make sure only trained responders attempt a technical rescue.
- Prohibit all non-essential personnel from the rescue site.
- Talk to the victim; determine the victim's condition, if possible.
- If you can reach the victim, check for vital signs, administer CPR, attempt to stop bleeding, and make the victim comfortable.

After an emergency, you must:

- Report fatalities and catastrophes to OSHA within eight hours.
- Report injuries requiring overnight hospitalization and medical treatment (other than first aid) to OSHA within 24 hours.
- Identify equipment that may have contributed to the emergency and put it out of service. Have a competent person examine equipment. If the equipment is damaged, repair or replace it. If the equipment caused the accident, determine how and why.
- Document in detail the cause of the emergency.
- Review emergency procedures. Determine how the procedures could be changed to prevent similar events; revise the procedures accordingly.

Click on the link to see a sample template for a written rescue and retrieval plan.

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5. Which course of action should be taken when a rescue at height emergency occurs?

a. Contact OSHA for direction
b. Don't upset the victim by initiating discussion
c. First responders should clear a path to the victim
d. Don't worry about documenting the accident workers are too stressed

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